

SE3090 Assignment 2 — Software Testing Report (Group SE_016, Family Veda) — Draft, 8 October 2026

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Reconciled on 8 October 2026 against retained repository evidence. No tests were rerun during this document review. Results dated 5 October are historical and do not certify the latest working tree.

SE3110 Quality Management in Software Engineering — Test Plan

Software Testing and Quality Evaluation of the SE3090 Integrated System

- **Project Title:** Family Veda — Longitudinal Family Health Context & Agentic Clinical Triage Platform
- **Academic Context:** SLIIT · BSc (Hons) in IT (Software Engineering) · Year 3 Semester 1 · 2026
- **Module Code:** SE3110 (Quality Management in Software Engineering)
- **Linked Assessment:** SE3090 Software Engineering Frameworks (Assignment 1, Group SE_016)
- **Document Version:** 1.0.0

- **Date:** October 2026
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1. Introduction & Objectives

1.1 Purpose

This Test Plan defines the comprehensive software testing strategy, scope, environment, tools, defect management lifecycle, and evaluation criteria for the **Family Veda** integrated platform. In strict accordance with the SE3110 specification, this testing plan evaluates the integrated system in controlled synthetic test environments across backend, database, web, mobile, agentic AI, and non-functional dimensions (performance and security).

1.2 Testing Objectives

1. **Verify Functional Correctness:** Ensure business workflows (authentication, consent, health record management, symptom triage, clinical approval) function accurately across web and mobile clients.
 2. **Validate Single-System Integration:** Verify that React Web and Flutter Mobile interface through the single shared ASP.NET Core Web API and PostgreSQL database without data drift or architectural divergence.
 3. **Enforce Clinical Safety Guardrails:** Deterministically evaluate the 10 Clinical Safety Rules (no direct diagnosis language, no drug dosing/prescriptions, strict human-in-the-loop doctor approval gate).
 4. **Quantify Non-Functional Resilience:** Measure system throughput, latency percentiles (p95, p99), and stability under concurrent user load using a proposed k6 run; the retained executed baseline used ApacheBench, while auditing API endpoints against OWASP security vulnerabilities.
 5. **Evaluate Agentic AI Subsystems:** Validate multi-agent orchestration, JSON-schema compliance, deterministic fallback mechanisms (Gemini → Groq → Safe failure), and prompt-injection resilience.
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2. System Architecture & Testing Scope

2.1 Technology Stack Under Test

- **Backend API:** ASP.NET Core 8.0 Web API (Clean Architecture: Domain, Application, Infrastructure, Api)
- **Database:** PostgreSQL 16 with Entity Framework Core (EF Core)
- **Web Frontend:** React 18, TypeScript, Vite, Redux Toolkit, Tailwind CSS (308 passing tests on 5 October; 343 in 57 files on 8 October)
- **Mobile Frontend:** Flutter 3.x, Dart, Riverpod state management (214 passing tests on 5 October; 256 on 8 October)
- **Agentic AI Subsystem:** 5 Application Agents (Context, Analysis, Extraction, Familial Risk, Safety/Validation) powered by Gemini 2.5 Flash / Groq LLMs and deterministic C# rule engines.

2.2 In-Scope Testing Areas (Per SE3110 Guidelines)

Testing Area	What Is Tested	Selected Frameworks & Tools
Backend / API Testing	Unit tests, service business logic, request validation (FluentValidation), controller endpoints, authentication/authorization (JWT, RBAC).	xUnit, Moq, WebApplicationFactory, FluentAssertions
Database Testing	Schema migrations, entity relational integrity, cascade deletes, unique constraints, JSONB column queries, transaction rollbacks.	xUnit, EF Core / PostgreSQL, Testcontainers for .NET
React Web Application	Component rendering, user form validation (Zod), protected routing (RouteGuard), async state management (Redux), error boundary handling.	Vitest, React Testing Library (RTL), jsdom (343 passing tests on 8 October; 308 on 5 October)
Flutter Mobile Testing	Mobile widget rendering, form input validation, Riverpod state providers, navigation flows, mock API interactions.	flutter_test, Mockito/mocktail
Integration & E2E Testing	Complete end-to-end clinical workflow: Patient complaint submission → Multi-agent triage → Doctor review & approval → Patient status notification.	Automated C# Workflow Integration Test / Playwright
Performance Testing (Mandatory)	API throughput, concurrency capacity, latency percentiles under sustained and ramped load (20–50 VUs) on Triage and Record endpoints.	k6 (proposed); ApacheBench (retained executed baseline)
Security Testing (Mandatory)	OWASP Top 10 API vulnerabilities, token forgery, consent-state bypass, broken object-level authorization (BOLA), injection vectors.	OWASP ZAP (Baseline scan), xUnit Security suites
Agentic AI Evaluation	Structured output schema validation, Tool allow-list enforcement (ToolDispatcher), Prompt injection defense, LLM fallback recovery.	xUnit deterministic test suites, System Prompt Test Harness

3. Team Member Allocation & Viva Demonstration Matrix

Individual viva performance represents **60 out of 100 marks**. The following allocation is supplied by the draft, not proof of actual work. Each student must confirm identity, ownership, authored tests and Git evidence. Responsibilities are proposed to map to student component ownership in SE3090 (docs/OWNERSHIP.tsv):

Member Ref	Student Name & IT Number	Owned Component	Primary Testing Tool / Role	Viva Demonstration Focus
S1	Samaranayaka S.G.V.S IT23544154	Family, Identity & Consent	Security Testing Lead (OWASP ZAP, xUnit Auth/Consent)	API Security Scan, JWT & Token lifecycle, Consent State Machine transitions, Tool Dispatcher allow-list denial.
S2	Fernando K.R.N IT24101875	Health Records & OCR Extraction	Database & Parsing Test Lead (xUnit DB constraints, Flutter Test)	PostgreSQL foreign key & cascade constraints, Lab OCR parser boundary cases, Corrupted file upload handling.
S3	Karunathilaka K.D.J.C IT24100551 (Group Leader)	Symptoms, Triage & Agent Orchestration	Performance & AI Evaluation Lead (k6 Load Testing, xUnit AI Suite)	k6 load test execution (p95 latency, concurrency metrics), Agent prompt injection resistance, LLM fallback recovery.
S4	W.M.S.S.B. Wasala IT24100559	Familial Risk & Clinical Approval	E2E & Safety Gate Test Lead (Playwright / Workflow Integration, xUnit)	Complete cross-component workflow trace, Deterministic Clinical Safety Rules 1–10 validation, Doctor Case Grant isolation.

4. Test Environment & Prerequisites

4.1 Environments

1. **Local Test Environment:**

- Current capture host: macOS; iPhone 18 Pro Max / iOS 27.0 simulator. Windows/Linux execution must be evidenced separately.
- Runtimes: .NET SDK 8.0.x, Node.js v20.x, Flutter 3.x, Python 3.11
- Database: PostgreSQL 16 (Local instance / Docker container)

2. **Continuous Integration (CI):**

- GitHub Actions workflow (.github/workflows/ci.yml) triggering on push and PR to main and develop.
- Automated jobs: Backend restore, build, test with XPlat coverage; Web lint, test with vitest; Mobile flutter analyze and test.

4.2 Test Data Isolation (Clinical Safety Rule 7)

- **Synthetic Data Only:** Under no circumstances are real patient identities, real NIC numbers, or real SLMC doctor registration numbers utilized.
- All test fixtures use pre-configured synthetic IDs, synthetic addresses, and standard mock lab data ranges.

5. Defect Management & Classification

5.1 Severity Levels

- **CRITICAL:** Application crash, database corruption, security breach/unauthorized data access, or violation of the 10 Clinical Safety Rules (e.g., AI output directly visible to patient without doctor approval).
- **HIGH:** Major functional defect, business workflow blocked, or unhandled 500 error on core API endpoints.
- **MEDIUM:** Validation error wording inconsistency, minor UI rendering quirk, or non-blocking edge-case failure.
- **LOW:** Cosmetic styling deviation, minor log format typo, or non-functional documentation gap.

5.2 Defect Lifecycle

Identified → Logged in Defect Register → Triageed & Assigned → Root Cause Diagnosed → Patch Developed → Unit/Integration Retest Verified → Closed.

6. Execution Schedule & Deliverables

Phase	Milestone	Target Output	Responsible
Phase 1	Test Planning & Architecture Setup	01_TEST_PLAN.md, environment setup	S3 (Janith)
Phase 2	Test Case Specification (Normal, Edge, Failure)	02_TEST_CASES_MASTER.md (41 supplied candidate cases; retained executed register separately)	S1, S2, S3, S4

Phase	Milestone	Target Output	Responsible
Phase 3 (planned)	Automated Script Implementation (k6, AI, Security)	scripts/testing/, new xUnit suites	S1, S2, S3, S4
Phase 4	Execution & Tool Evidence Collection	Test logs, coverage reports, k6 graphs, ZAP scans	All members
Phase 5	Defect Logging & Retesting Documentation	03_DEFECT_LOG.md (12 retained defect entries; supplied additional claims require evidence)	All members
Phase 6	Consolidated Software Testing Report Compilation	Markdown report package (user-requested format)	Group Leader (S3)

Evidence-led execution and completion gates

3. Non-Functional Test Selection

Performance and security testing are mandatory for this assessment. Table 2 records which non-functional types were selected, and why the others were not.

Table 2 – Non-Functional Test Selection

Non-functional type	Selected	Justification
Performance (load profile)	Yes — required	Confirms that the read paths both portals depend on complete without errors under concurrency, and shows which are slowest. Limited to read endpoints and a local environment (Section 7).
Security	Yes — required	The system handles health-related records and cross-profile access, so access control, token handling, dependency vulnerabilities and malformed-input handling carry the highest risk. Covered by automated access-control tests, dependency audits, static analysis and an OWASP ZAP scan of the API (Section 6).
Reliability and recovery	Yes	A triage case must never be lost or silently replayed. Worker-restart recovery, provider-failure fallback and safe-failure paths are covered by automated tests.
Compatibility	Partly	The mobile application was built and run on an Android API 36 emulator only. No physical device and no iOS build were tested.
Stress testing	No	Free-tier hosting imposes its own limits, so a stress result would describe the hosting plan rather than the application.
Usability and accessibility	No	No tool-based usability or accessibility run (for example Lighthouse or axe) was executed. Responsive layout was checked manually during development, which does not count as tool-based evidence.

Entry / exit criteria and risk controls

Entry: record commit and dirty state, runtimes, disposable local PostgreSQL target, synthetic fixtures and tool versions. Docker must be healthy for Testcontainers. Keep credentials outside Markdown and evidence.

Exit: execute selected normal/invalid/boundary/failure tests; retain command, date, revision, logs and coverage; document failures and independent retests; complete one correlated integrated workflow plus required performance and ZAP evidence. Any unmet gate is reported as Partial or Not Run. No overall 100% coverage is calculated by averaging unrelated suites.

Prioritise the doctor approval gate, consent and case grants, emergency referral and fail-safe handling. Performance/active scanning targets disposable local environments. Proposed k6 thresholds (p95 <500ms, errors <1%) are acceptance targets requiring a recorded agreed workload, not existing results.

See [completion checklist](#), [cases](#), [defects](#) and [execution summary](#).

Master test case register

Reconciled on 8 October 2026 against retained repository evidence. No tests were rerun during this document review. Results dated 5 October are historical and do not certify the latest working tree.

5. Test Case Document

Table 4 is the test case document. Each case names the feature, its type (normal, invalid, boundary or failure), the preconditions and steps, the expected and actual result, and the status. Cases TC-01 to TC-26 are individual automated tests, identified by their test method so that they can be rerun; cases A2-* are the wider integrated, non-functional and device cases executed on 2026-09-28. Automated cases were last executed on 2026-10-05 as part of the suites in Table 3.

Table 4 – Test Case Document

ID	Feature	Type	Preconditions and steps / input	Expected result	Actual result	Status
TC-01	Registration (S1)	Normal	Submit a complete Family Head registration to the validator — ValidFamilyHead_Passes	Validation passes	Passed	Pass
TC-02	Registration (S1)	Invalid	Submit a Family Head registration with no family name — FamilyHead_RequiresFamilyName	Validation error on family name	Error returned	Pass
TC-03	Registration (S1)	Invalid	Submit mismatched password and confirmation — ConfirmPassword_MustMatch	Validation error	Error returned	Pass
TC-04	Registration (S1)	Boundary	Submit a date of birth below the adult threshold, and a future date — DateOfBirth_RequiresAdult, DateOfBirth_RejectsFuture	Both rejected	Both rejected	Pass
TC-05	Consent (S1)	Boundary	Guardian-granted consent for a member who turns eighteen — RequiresReaffirmation_WhenGuardianGrantedAndMemberTurnsEighteen_ReturnsTrue	Reaffirmation required	Reaffirmation required	Pass
TC-06	Authentication (S1)	Failure	Submit the same refresh token concurrently — RefreshToken_IsSingleUse_WhenSubmittedConcurrently	Only one request succeeds	One succeeded	Pass
TC-07	Tool dispatch (S1)	Failure	An agent requests a tool outside its allow-list — DeniedTool_IsPersisted_AndThrowsHardError	Denial persisted; processing stops	Denial persisted; error raised	Pass
TC-08	Tool dispatch (S1)	Normal	Familial Risk Agent reads relatives' flags — FamilialRisk_ReadsOnlyBiologicalRelativesWithCurrentConsent	Only consented biological relatives returned	As expected	Pass
TC-09	Lab extraction (S2)	Normal	Parse a recognised report table — ParseValues_ExtractsStructuredRows_AndIgnoresFreeText	Structured rows extracted; free text ignored	As expected	Pass

ID	Feature	Type	Preconditions and steps / input	Expected result	Actual result	Status
TC-10	Lab extraction (S2)	Invalid	Parse text containing appointment times — ParseValues_DoesNotTreatAppointmentTimesAsLabResults	No lab value created	None created	Pass
TC-11	Lab extraction (S2)	Invalid	Parse table rows with malformed fields or no recognised header — ParseValues_TableRows_UseCurrentValueAndRejectMalformedFields, ParseValues_DoesNotParseTableRowsWithoutRecognizedHeader	Malformed rows rejected	Rejected	Pass
TC-12	Lab extraction (S2)	Failure	Extraction recognises zero rows — ExtractAsync_WhenRecognizedRowsAreZero_FailsWithoutWritingValuesOrFlags	Fails without writing values or flags	Nothing written	Pass
TC-13	Lab extraction (S2)	Failure	Re-extract a report that already holds manually confirmed data — ExtractAsync_WhenReportContainsManuallyConfirmedData_RejectsAndPreservesConfirmedRows	Rejected; confirmed rows preserved	Preserved	Pass
TC-14	Agent orchestration (S3)	Failure	An agent returns output that violates its schema — InvalidAgentSchema_FailsSafe_AndStopsRemainingAgents	Case fails safe; later agents do not run	Failed safe	Pass
TC-15	Emergency gate (S3)	Failure	Emergency wording appears in free-text notes — EmergencyGate_AlsoChecksFreeTextNotes_BeforeEveryLlmAgent	Halt before every LLM agent	Halted	Pass
TC-16	LLM client (S3)	Failure	Provider returns rate-limit responses twice — RateLimitedTwice_FailsClosed	Client fails closed	Failed closed	Pass
TC-17	LLM client (S3)	Invalid	Provider returns a schema violation once — SchemaViolation_RetrievesOnce_ThenReturnsValidOutput	One retry, then valid output	As expected	Pass
TC-18	Worker recovery (S3)	Boundary	More queued cases than channel capacity at start-up — Recovery_ReturnsMoreThanChannelCapacity_WithoutBlockingStartup	Start-up not blocked	Not blocked	Pass
TC-19	Case grant (S4)	Boundary	Grant expired or exactly at its expiry time — HasAccess_WhenGrantIsExpiredOrAtExpiryBoundary_ReturnsFalse	Access denied	Denied	Pass
TC-20	Case grant (S4)	Invalid	Grant was revoked — HasAccess_WhenGrantWasRevoked_ReturnsFalse	Access denied	Denied	Pass
TC-21	Safety validation (S4)	Failure	Emergency red flag present — Validate_WhenEmergencyRedFlagExists_HaltsBeforeLlmAndReturnsReferralOnly	Referral only; no AI output	Referral only	Pass
TC-22	Safety validation (S4)	Invalid	Evidence contains prohibited content — Validate_WhenLabEvidenceContainsMedicationContent_StillBlocksContent	Content blocked	Blocked	Pass

ID	Feature	Type	Preconditions and steps / input	Expected result	Actual result	Status
TC-23	Approval gate (S4)	Normal	Complete synthetic golden case — SyntheticGoldenCase_RequiresApprovalBeforeFamilyCanReadGuidance	Guidance unreadable before approval, readable after	As expected	Pass
TC-24	Doctor verification (S4)	Invalid	Unverified doctor calls clinical routes — PendingDoctor_IsForbiddenFromEveryClinicalQueueAndDecisionSurface	Forbidden on every route	Forbidden	Pass
A2-API-01	Three-role API journey	Normal	Synthetic Head, Adult Member and Doctor against local API and PostgreSQL 16; run scripts/e2e/synthetic_portal_journey.py	Journey completes; Head denied an adult's records	Join by code, doctor acceptance, 45-minute booking, dashboards and notifications passed; Head received 404	Pass
A2-API-02	Doctor acceptance	Failure (regression)	Fixture with an already assigned doctor and a legacy pending request; accept the request	No duplicate assignment	HTTP 200, one assignment, request Accepted	Pass
A2-DB-02	Doctor concurrency constraints	Boundary	Competing writes for pending requests and active primary assignments; concurrent accept and decline	At most one of each; one decision with matching audit	One write won each pair; API returned one 200 and one 409 with one audit row	Pass
A2-DB-03	Idempotent migration	Normal	Fresh PostgreSQL 16; apply the first two migrations, then the doctor-constraint script twice	Both runs succeed; one history row per migration	Both runs exited 0; three distinct migration IDs	Pass
A2-DB-04	Populated migration and rollback	Failure	Legacy database with history; conflicting legacy database; repeated assignment after upgrade	Rows preserved; conflicting upgrade rejected atomically; unsafe rollback refused	Confirmed by integration tests	Pass
A2-MOB-02	Android appointment booking	Normal	Synthetic Head on Android API 36 emulator with local API; select 45 minutes and 10:00 AM; submit	Same local time shown in the appointment list	Stored duration_minutes=45; list showed 10:00 AM after the time-zone fix	Pass
A2-MOB-03	Hosted-API Android APK	Normal	Build with the production API URL; install on Android API 36 emulator; sign in as synthetic Head	APK builds, installs and reaches the dashboard	Built, installed, signed in; dashboard, appointments and notifications loaded. Doctor-approved guidance journey on the hosted environment not run	Partial
A2-E2E-01	Complete integrated workflow	Normal and failure	Flutter → API, database and agents → React approval → Flutter result	Golden case and safe-failure case demonstrated across both clients	Connected Flutter/web run on 8 Oct found two High defects (fixed, PR #176). Hosted-agent golden journey then passed 11/11 checks through the API; visual repeat after the fix not captured	Pass (API), Partial (visual)
A2-PERF-01	Performance	Normal	ApacheBench, 1,000 requests at concurrency 25 on each of 14 read endpoints (14,000 requests), disposable local API and PostgreSQL 16; run scripts/e2e/local_load_profile.py	Zero failed or non-2xx responses on every endpoint	14,000 completed, 0 failed, 0 non-2xx; slowest endpoint 88 ms mean and 267 ms p99 (Table 7)	Pass (local)
A2-SEC-01	Access control	Invalid	Unauthenticated GET on three protected live routes; preflight from an untrusted origin	401 on each route; no allow-origin header	401 returned three times; no allow-origin header	Pass (scoped)

ID	Feature	Type	Preconditions and steps / input	Expected result	Actual result	Status
A2-SEC-02	Web dependency audit	Normal	npm audit --omit=dev --audit-level=high	No known vulnerabilities	found 0 vulnerabilities (2026-10-05)	Pass
A2-SEC-03	API dependency audit	Normal	dotnet list package --vulnerable --include-transitive	No vulnerable packages	None reported (2026-10-05)	Pass
A2-SEC-04	Dynamic security scan	Normal and invalid	OWASP ZAP API scan over the OpenAPI definition with a synthetic Family Head token, disposable local API	No high- or medium-risk alert	0 failed; 4 low-risk warnings on first run, 2 after fixes (D-011, D-012)	Pass
TC-25	Request hardening (S1)	Invalid	Send a query string containing a NUL character — NulCharacterInQueryString_IsRejectedAsBadRequest_NotServerError	HTTP 400 problem details, not HTTP 500	HTTP 400	Pass
TC-26	Request hardening (S1)	Normal	Request the health and an API route — EveryResponse_TellsBrowsersNotToSniffContentType	X-Content-Type-Options: nosniff on each	Present	Pass
A2-WEB-01	Vitals panel (web)	Normal	Run VitalsPanel.test.tsx on 2026-10-05	All tests pass	First run: two of four failed (D-008). Retest after the tests were aligned with the redesigned panel: three of three passed	Pass (after retest)
A2-WEB-02	Web quality gate	Normal	npm run lint and npm run build on 2026-10-05	Both exit 0	First run: lint exit 1, build exit 2 (D-009). Retest after fix: both exit 0	Pass (after retest)

Of the 41 cases, 39 are Pass (two with stated scope qualifications and two after a same-day fix and retest) and 2 are Partial. No case remains failed.

Supplied candidate scenarios — not verified executions

The supplied file states 44 cases but contains 41 case rows. These are retained as planning candidates; their original PASS/actual claims are not accepted without a matching test method, command, revision and log. Endpoint names, response codes, bounds and safety expectations must be checked against current code before execution. The supplied example password is intentionally omitted.

Candidate ID	Category	Layer	Scenario	Status
TC-S1-001	Normal	Backend API	User registration with valid synthetic credentials	Not verified; planned only
TC-S1-002	Invalid	Backend API	Register with invalid email format	Not verified; planned only
TC-S1-003	Boundary	Web (Vitest)	Password length boundary validation	Not verified; planned only
TC-S1-004	Failure	Backend API	Login with incorrect password	Not verified; planned only
TC-S1-005	Normal	Backend Unit	Consent State Machine grant transition	Not verified; planned only
TC-S1-006	Invalid	Backend Unit	Consent Revocation invalid transition	Not verified; planned only

Candidate ID	Category	Layer	Scenario	Status
TC-S1-007	Security	Backend API	RBAC unauthorized endpoint access	Not verified; planned only
TC-S1-008	Security	Backend Unit	Tool Dispatcher unauthorized tool denial	Not verified; planned only
TC-S1-009	Normal	Mobile (Flutter)	Login screen widget rendering and successful submission	Not verified; planned only
TC-S1-010	Boundary	Database	Unique constraint on User Email	Not verified; planned only
TC-S2-001	Normal	Backend API	Upsert manual health record with vitals	Not verified; planned only
TC-S2-002	Invalid	Backend API	Record vitals with negative numbers	Not verified; planned only
TC-S2-003	Boundary	Backend Unit	Lab extraction parser upper reference boundary	Not verified; planned only
TC-S2-004	Failure	Backend Unit	Lab report parser on unreadable / corrupted file	Not verified; planned only
TC-S2-005	Normal	Web (Vitest)	Recorded Range Visual component indicator	Not verified; planned only
TC-S2-006	Boundary	Web (Vitest)	Empty record list rendering	Not verified; planned only
TC-S2-007	Normal	Mobile (Flutter)	Lab report file picker and upload preview	Not verified; planned only
TC-S2-008	Database	Database	Foreign key cascade deletion rule	Not verified; planned only
TC-S2-009	Failure	Backend Unit	Lab extraction safety with synthetic clinical override	Not verified; planned only
TC-S2-010	Boundary	Database	JSONB LabValues serialization bounds	Not verified; planned only
TC-S3-001	Normal	Backend API	Submit symptom triage complaint	Not verified; planned only
TC-S3-002	Invalid	Backend API	Submit empty complaint description	Not verified; planned only
TC-S3-003	Boundary	Backend API	Maximum symptom character limit check	Not verified; planned only
TC-S3-004	Failure	AI Evaluation	LLM Provider Timeout & Fallback Execution	Not verified; planned only
TC-S3-005	Failure	AI Evaluation	All LLM Providers Offline (Safe Failure)	Not verified; planned only
TC-S3-006	Security	AI Evaluation	Prompt Injection & Jailbreak Defense	Not verified; planned only
TC-S3-007	Normal	Web (Vitest)	Triage Agent Pipeline visual stages	Not verified; planned only
TC-S3-008	Normal	Mobile (Flutter)	Submit complaint screen chip selector	Not verified; planned only
TC-S3-009	Performance	k6 Load Test	Triage API concurrency load test	Not verified; planned only
TC-S3-010	Normal	Backend Unit	Case SLA Background Processor	Not verified; planned only
TC-S3-011	Normal	Backend Unit	Structured JSON-Schema Output Validation	Not verified; planned only
TC-S4-001	Normal	Backend API	Verified Doctor reviews and approves case	Not verified; planned only
TC-S4-002	Security	Backend API	Doctor attempts to approve ungranted case	Not verified; planned only
TC-S4-003	Failure	Backend Unit	Clinical Safety Rule 1: Zero Diagnosis Language	Not verified; planned only
TC-S4-004	Failure	Backend Unit	Clinical Safety Rule 6: Prohibited Drug Dosing	Not verified; planned only
TC-S4-005	Normal	Backend Unit	Clinical Safety Rule 10: Chest Pain Emergency Bypass	Not verified; planned only
TC-S4-006	Normal	Backend Unit	Familial Risk Pedigree Screening Indication	Not verified; planned only

Candidate ID	Category	Layer	Scenario	Status
TC-S4-007	Normal	Web (Vitest)	Doctor Approvals Page card and decision buttons	Not verified; planned only
TC-S4-008	Boundary	Web (Vitest)	Doctor Approval Notes character boundary	Not verified; planned only
TC-S4-009	Normal	E2E Integration	Complete Cross-Platform Clinical Lifecycle Workflow	Not verified; planned only
TC-S4-010	Normal	Web (Vitest)	Safety Rules badge display check	Not verified; planned only

For execution, use the full preconditions/steps/expected/actual/status/evidence template in the [complete pack](#). Do not reuse guessed `/api/...` routes where the implementation uses `/api/v1/...`. Classify parser extraction against report-printed reference ranges; do not add patient diagnosis, dosing or automatic clinical advice.

Test execution summary

Reconciled on 8 October 2026 against retained repository evidence. No tests were rerun during this document review. Results dated 5 October are historical and do not certify the latest working tree.

4. Automated Suite Results

Table 3 reports the retained 2026-09-28 execution alongside the final local rerun on 2026-10-05 at commit 613bf7e on develop. The rerun output is retained under evidence/2026-10-05/. Results are reported as observed, including failures.

Table 3 – Automated Test Suite Results

Suite	Tool	Retained run 2026-09-28	Rerun 2026-10-05 @ 613bf7e	Line coverage (2026-10-05)	Evidence file
Backend unit	xUnit + Moq	91 / 91 passed	337 / 337 passed	20.2% overall; Domain 72.4%, Application 76.0%	backend-unit.txt
PostgreSQL integration	xUnit + Testcontainers (PostgreSQL 16)	11 / 11 passed	25 / 25 passed; 29 / 29 after the four request-hardening tests were added	69.1% overall; Infrastructure 71.1%, Api 59.0%	backend-integration.txt, backend-integration-final.txt
Web tests	Vitest + React Testing Library	41 / 41 passed	First run 306 / 308 (2 failed); retest after fixes 308 / 308 passed (52 files)	69.6% lines; 67.5% statements; 62.4% branches	web-tests.txt, web-tests-retest.txt, web-coverage.txt
Web lint and build	ESLint; tsc -b + Vite build	Passed	First run failed (lint exit 1, build exit 2); retest after fix: both exit 0	—	web-lint-build.txt, web-lint-build-retest.txt
Flutter	flutter analyze; flutter_test	No issues; 69 / 69 passed	No issues; 214 / 214 passed	76.4%	flutter-tests.txt
Dependency audits	npm audit --omit=dev; dotnet list package --vulnerable	0 vulnerabilities; none reported	0 vulnerabilities; none reported	—	dependency-audits.txt

Three points qualify these figures. First, the first run at 613bf7e exposed two defects, both from same-day changes to the records pages: two obsolete tests in web/src/pages/records/VitalsPanel.test.tsx (D-008) and an unused Link import in web/src/pages/records/RecordsPage.tsx that failed lint and the production build (D-009). Both were fixed and retested the same day; the final suite has 308 tests because one obsolete test was removed and one guard test was added with D-010 (Table 8). Web coverage was captured in a separate run with fewer workers and a longer test timeout, because coverage-instrumented runs timed out when the development machine was heavily loaded. Second, backend coverage is reported per run and has not been merged: the unit run covers the Domain and Application layers well but barely touches controllers and persistence, which the integration run covers instead. The backend line totals also include generated EF Core migration code, which lowers the overall percentage. Third, the growth from the retained run to the rerun (for example 91 to 337 unit tests) reflects tests added with the features delivered between the two dates.

Non-functional results

6. Security Testing

Table 6 lists the security checks for which execution evidence exists. Each is automated and tool-based; together they are still not a substitute for an independent penetration test.

Table 6 – Security Checks

ID	Check	Expected	Actual	Status
SEC-1	Guidance requested before doctor approval (integration test)	Family read of unapproved guidance denied	HTTP 404 before approval; guidance returned only after allow-listed approval	Pass
SEC-2	Action by a doctor whose verification is pending (integration test)	Denied	Pending-doctor denial confirmed	Pass
SEC-3	Refresh-token reuse (integration test)	Refresh token usable once only	Single-use behaviour confirmed	Pass
SEC-4	Tool dispatch, case grants and consent (11 unit tests)	Unauthorised tool use, stale grants and revoked consent are denied	11 unit tests passed	Pass
SEC-5	Unauthenticated GET to /api/v1/notifications, /api/v1/dashboard/family, /api/v1/dashboard/doctor on the live API	HTTP 401	All three returned HTTP 401	Pass
SEC-6	CORS preflight from https://untrusted.example.invalid	No allow-origin header	HTTP 204 without Access-Control-Allow-Origin	Pass
SEC-7	npm audit --omit=dev --audit-level=high (web production dependencies)	No known vulnerabilities	found 0 vulnerabilities (2026-09-28 and 2026-10-05)	Pass
SEC-8	dotnet list package --vulnerable --include-transitive (API)	No known vulnerable packages	None reported (2026-09-28 and 2026-10-05)	Pass
SEC-9	OWASP ZAP API scan of all endpoints in the OpenAPI definition, authenticated as a synthetic Family Head, against a disposable local API	No high- or medium-risk alert	First run: 116 rules passed, 0 failed, 4 low-risk warnings. Retest after fixes: 117 passed, 0 failed, 2 low-risk warnings (Section 6.2)	Pass
SEC-10	Query string carrying a NUL character (integration test)	HTTP 400, not a server error	HTTP 400 with a problem-details body	Pass
SEC-11	X-Content-Type-Options header on API responses (integration test)	nosniff on every response	Present on the health and API routes tested	Pass

Checks SEC-1 to SEC-3 are the three integration tests, and SEC-4 the eleven unit tests, that together form the focused xUnit run of A2-SEC-01 (3 + 11 tests). A further privacy check, that a Family Head receives HTTP 404 on an adult member's records, was part of the synthetic API journey A2-API-01 in Table 4. SEC-10 and SEC-11 are the regression tests in RequestHardeningTests added after the scan.

6.1 Mapping to OWASP

The executed checks map to the OWASP Top Ten 2021 categories (OWASP Foundation, 2021) as follows.

- **A01 Broken Access Control.** SEC-1, SEC-2, SEC-4, SEC-5 and the Head-404 privacy check address object-level and function-level authorisation, grant expiry and consent revocation.
- **A02 Cryptographic Failures and A07 Identification and Authentication Failures.** SEC-3 covers refresh-token single use. Password hashing and token lifetimes are design controls described in Chapter 3; they are not separately tested here beyond the checks listed.
- **A03 Injection.** SEC-9 exercised every endpoint with the scanner's injection rules (SQL injection, command injection, server-side template injection, path traversal and others) and raised no injection alert. SEC-10 covers the one malformed-input fault it did find.
- **A05 Security Misconfiguration.** SEC-6 covers CORS handling for an untrusted origin; SEC-9 and SEC-11 cover response security headers.
- **A06 Vulnerable and Outdated Components.** SEC-7 and SEC-8 are dependency audits; CodeQL and Dependabot provide continuing coverage.

For the agentic subsystem, the controls are relevant to categories in the OWASP Top 10 for large language model applications (OWASP Foundation, 2025), in particular prompt injection and excessive agency. Indirect prompt injection through untrusted content is a recognised risk for LLM-integrated applications (Greshake et al., 2023). The relevant mitigations are structural: agents hold no database credentials, tool access is limited to a per-agent allow-list enforced at dispatch (`ToolRegistryTests`, `ToolDispatcherTests`, SEC-4), safety validation is deterministic rather than an LLM judgement, and no output reaches a patient without doctor approval (SEC-1). OCR text is treated as untrusted input. This mapping describes which controls address which categories; it does not claim that every category was tested.

6.2 Dynamic Scan with OWASP ZAP

A dynamic scan was run with OWASP ZAP 2.17.0 using its API scan, which imports the OpenAPI definition published at `/swagger/v1/swagger.json` and then both passively inspects and actively attacks every documented endpoint. The scanner was given a bearer token for a synthetic Family Head so that it could reach authenticated routes. The target was a disposable local API with its own PostgreSQL 16 database and synthetic seed data; the hosted service and its database were deliberately not scanned.

The first run passed 116 rules, failed none and raised four low-risk warnings. No high- or medium-risk alert was raised in either run. The four warnings and their outcome were:

1. **A server error on malformed input.** GET `/api/v1/doctors/directory?search=%00` returned HTTP 500, because PostgreSQL cannot store a NUL character in text. This was a real defect (D-011): the client received a server error for what is a client error. It is now rejected with HTTP 400 before reaching the database.
2. **X-Content-Type-Options header missing.** A real hardening gap (D-012). Every response now carries `nosniff`.
3. **Cross-Origin-Resource-Policy header missing.** Accepted, not changed. The web client is served from a different site from the API, so a restrictive value risks blocking legitimate requests, and the API returns JSON to authenticated callers rather than embeddable resources. It is recorded as a limitation.
4. **Unexpected content type on /swagger/.** A false positive: the Swagger UI page is HTML by design, while the scanner expects JSON from an API.

After the two fixes the scan was repeated: 117 rules passed, none failed, and only warnings 3 and 4 remained. The summary and both full reports are retained in `evidence/2026-10-05/security-zap-summary.txt`, `zap-first-run/` and `zap-retest/`.

The scan has limits. It ran with one role's token, so it did not test whether a Doctor or Clinic Admin token can reach another role's data; that property is covered by the access-control tests SEC-1 to SEC-5 instead. It tested the API only, not the web or mobile clients. The evaluation as a whole remains **automated testing, not a penetration test**: session-management attacks, business-logic abuse and independent review were not performed (see Section 10).

7. Performance Testing

Performance was measured with ApacheBench against a disposable local API and PostgreSQL 16 database loaded with the synthetic seed data. The script `scripts/e2e/local_load_profile.py` signs in as a synthetic Family Head and a synthetic Doctor, then sends 1,000 requests at a concurrency of 25 to each of fourteen read endpoints: 14,000 requests in total. It refuses any non-local host and fails if any request fails or returns a non-2xx status. Table 7 reports the result; the raw output is retained in `evidence/2026-10-05/performance-load-profile.txt`.

Table 7 – Performance Baseline

Endpoint	Completed	Failed	Non-2xx	Requests/s	Mean (ms)	p50 (ms)	p95 (ms)	p99 (ms)	Max (ms)
Health check (anonymous)	1000	0	0	21,439	1.2	1	2	2	2
Family: own family	1000	0	0	2,660	9.4	7	12	79	89
Family: dashboard	1000	0	0	284	88.2	67	197	267	384
Family: notifications	1000	0	0	2,063	12.1	7	34	93	214
Family: appointments	1000	0	0	639	39.1	33	73	115	191
Family: health records	1000	0	0	731	34.2	27	72	115	181
Family: vitals	1000	0	0	1,041	24.0	19	55	120	184
Family: lab reports	1000	0	0	938	26.6	23	47	81	114
Family: triage cases	1000	0	0	1,289	19.4	16	34	55	94
Family: doctor directory	1000	0	0	1,977	12.6	10	27	37	64
Doctor: dashboard	1000	0	0	452	55.3	50	88	121	162
Doctor: my cases	1000	0	0	593	42.1	37	78	125	242
Doctor: case pool	1000	0	0	530	47.1	42	87	118	164
Doctor: appointments	1000	0	0	639	39.1	21	127	560	570

Mean is the mean time per request at the stated concurrency. All fourteen endpoints completed every request with no failed and no non-2xx responses.

Interpretation. Three observations follow from Table 7. First, the two dashboard endpoints are the slowest (88 ms and 55 ms mean): each aggregates several queries into one response, so they are the first place to optimise, for example by combining queries or caching the summary briefly. Second, simple list endpoints stay between 9 ms and 47 ms mean, and every endpoint keeps its 95th percentile under 200 ms. Third, the doctor appointments endpoint has a long tail (560 ms at the 99th percentile against a 21 ms median), which points to occasional contention rather than a consistently slow query and deserves a profiling pass.

The result should be read conservatively. The run was local, with no network latency, a small synthetic data set and a development build. It covers read paths only: no write load and no mixed workload. It says nothing about the hosted environment, where the free-tier API instance sleeps when idle and adds a cold-start delay to the first request, nor about the latency of the agent pipeline, which is dominated by hosted LLM response times (provider timeouts are 45 and 30 seconds). An earlier single-endpoint baseline of 200 requests on 2026-09-28 (1.8 ms mean, zero failures) is retained in the Assignment 2 execution record.

Completion assessment

The four main suite logs show $337 + 29 + 308 + 214 = \mathbf{888 \text{ historical passing tests}}$. This is a suite total, not 888 independent report cases; AI/security subsets must not be added again. Coverage scopes are separate and no 84.8% overall coverage is supported. Full connected client E2E remains Partial. Performance is a local read-only ApacheBench baseline, not a k6 triage-write measurement or production SLA certification. ZAP retest retains two low-risk warnings and used one role token.

135 current light-mode screenshots document navigation only: [gallery](#). Report is not final submission-ready until the [completion checklist](#) is resolved.

Latest execution update

8 October local working-tree rerun: **430 backend + 30 PostgreSQL integration + 343 React + 256 Flutter = 1,059 passed**. Web lint/build and Flutter analyze also passed. [Commands, logs, TRX and limits](#). Historical coverage remains separate.

Current coverage and non-functional evidence

[8 October coverage, performance and ZAP results](#) now includes actual artifacts. Coverage: unit24.71%, integration64.28%, React71.94%, Flutter75.75% lines (scopes differ). ApacheBench14,000 requests completed without failed/non-2xx responses. ZAP0 High/Medium,2 Low warning types retained. Connected UI E2E status is recorded separately.

Connected client execution outcome

[8 October connected Flutter/web E2E](#) was executed against isolated local infrastructure. Fresh case0021 failed safely because a hosted agent was unavailable. Seeded case0003 approval propagated to Flutter, but the run exposed High internal-note privacy and stale-delay navigation defects. Overall status remains Failed/Partial, not Pass; fixes/retests are pending.

Current execution evidence — 8 October 2026

Local HEAD: 3bb6c192a379f73eccbfea2745f6fee26fec87e4. Working tree contained uncommitted changes; see [snapshot](#). No pull/reset was performed. GitHub develop showed a newer merge, so this run does not certify the later remote state.

Actual execution

Check	Command	Actual result	Evidence
Backend unit	dotnet test backend/tests/UnitTests/FamilyVeda.UnitTests.csproj --no-restore	430 passed; 0 failed; 0 skipped	backend-unit.txt
PostgreSQL integration	dotnet test backend/tests/IntegrationTests/FamilyVeda.IntegrationTests.csproj --no-restore	30 passed; 0 failed; 0 skipped	backend-integration.txt
React	cd web && npx vitest run --maxWorkers=2	57 files; 343 passed	web-tests.txt
Flutter	cd mobile && flutter test --no-pub	256 passed	flutter-tests.txt
Web lint	cd web && npm run lint	Exit 0	web-lint.txt
Web build	cd web && npm run build	Exit 0; bundle-size warning retained	web-build.txt
Flutter analyze	cd mobile && flutter analyze --no-pub	No issues found; exit 0	flutter-analyze.txt

All commands exited 0. Total across the four suites: **1,059 passing tests**. Coverage was collected in a separate 8 October run; see [08 CURRENT NONFUNCTIONAL RESULTS.md](#). xUnit runs also retained TRX files. [Runtime versions](#).

Selected tool-based safety and defect retests

The following named tests are independently recorded in the current integration TRX, which provides per-test outcomes. This is API/database proof, not a connected visual client workflow.

Test	Current outcome
FamilyVeda.IntegrationTests.RequestHardeningTests.EveryResponse_TellsBrowsersNotToSniffContentType(path: "/health")	Passed
FamilyVeda.IntegrationTests.GoldenCaseFlowTests.InvalidAgentSchema_FailsSafe_AndKeepsFamilyGuidanceUnavailable	Passed
FamilyVeda.IntegrationTests.GoldenCaseFlowTests.DoctorOriginalReport_UsesScopedAuthorizationAndSafeHeaders	Passed
FamilyVeda.IntegrationTests.AuthAndPatientFlowTests.PendingDoctor_IsForbiddenFromEveryClinicalQueueAndDecisionSurface	Passed
FamilyVeda.IntegrationTests.RequestHardeningTests.EveryResponse_TellsBrowsersNotToSniffContentType(path: "/api/v1/doctors/directory")	Passed
FamilyVeda.IntegrationTests.GoldenCaseFlowTests.SyntheticGoldenCase_RequiresApprovalBeforeFamilyCanReadGuidance	Passed
FamilyVeda.IntegrationTests.RequestHardeningTests.NulCharacterInQueryString_IsRejectedAsBadRequest_NotServerError	Passed
FamilyVeda.IntegrationTests.GoldenCaseFlowTests.PatientWithdrawal_RacesFullDoctorReview_ExactlyOneCanWin	Passed

Test	Current outcome
FamilyVeda.IntegrationTests.AuthAndPatientFlowTests.RefreshToken_IsSingleUse_WhenSubmittedConcurrently	Passed

[Integration TRX](#) · [unit TRX](#).

Evidence status

Required evidence	Actual value / evidence
Current coverage per suite	Collected 8 October; document 08 ; evidence/2026-10-08-coverage/
Load test execution	ApacheBench, 14,000 requests, 0 failed; document 08
New ZAP execution	117 PASS / 0 FAIL / 2 WARN; evidence/2026-10-08-zap/
Connected Flutter → web doctor → Flutter journey	Executed, Failed/Partial; document 09
Student personally executed tool demonstrations	
Student CLEAR declarations	

The two blank rows are for each student to complete; blank is not a Pass or an exemption. The fix made afterwards for the two connected-client defects is recorded in [03_DEFECT_LOG.md](#).

Verification qualifications

Golden workflow tests use deterministic stub agents; they prove approval/safe-failure API invariants, not hosted Gemini/Groq quality or a connected client demonstration. Flutter output contains missed-tap warnings despite the suite passing; retain and review those warnings rather than describe the run as warning-free. TRX files are present in this local evidence folder but may be Git-ignored; include them explicitly in the submission package if packaging it later.

Current coverage, performance and security — 8 October 2026

Coverage results

Suite	Passing tests	Line coverage	Branch coverage
Backend unit	430	11,191 / 45,297 — 24.71%	3,703 / 6,767 — 54.72%
PostgreSQL integration	30	29,118 / 45,297 — 64.28%	939 / 6,767 — 13.88%
React	343	3,619 / 5,030 — 71.94%	3,413 / 5,297 — 64.43%
Flutter	256	8,632 / 11,396 — 75.75%	Not instrumented

Backend scope includes Api, Application, Domain and Infrastructure, including generated migrations. Application unit coverage is 672/873 lines (76.98%); this does not establish an 80% touched-service gate. Unit/integration coverage is not merged or averaged. Flutter covers 122 files. Web also records functions 66.52% and statements 69.80%. All 1,059 tests pass; Flutter missed-tap warnings remain. The first web coverage attempt had an unretained log after output-folder cleanup; the preserved rerun passed.

Coverage artifact index

- [backend-integration/891c168e-3283-4eb7-b4db-17a635143995/coverage.cobertura.xml](#)
- [backend-unit/0a87f8e9-51d8-4119-9cb9-aea69e654556/coverage.cobertura.xml](#)
- [coverage-summary-calculated.json](#)
- [mobile/lcov.info](#)
- [web/report/assets/index.html](#)
- [web/report/clover.xml](#)
- [web/report/components/auth/index.html](#)
- [web/report/components/layout/index.html](#)
- [web/report/components/records/index.html](#)
- [web/report/components/shared/index.html](#)
- [web/report/coverage-final.json](#)
- [web/report/index.html](#)
- [web/report/pages/admin/index.html](#)
- [web/report/pages/audit/index.html](#)
- [web/report/pages/auth/index.html](#)
- [web/report/pages/dashboard/index.html](#)
- [web/report/pages/doctor/calendar/index.html](#)
- [web/report/pages/doctor/index.html](#)

- [web/report/pages/family/index.html](#)
- [web/report/pages/notifications/index.html](#)
- [web/report/pages/profile/index.html](#)
- [web/report/pages/public/index.html](#)
- [web/report/pages/records/index.html](#)
- [web/report/pages/system/index.html](#)
- [web/report/pages/triage/index.html](#)
- [web/report/routes/index.html](#)
- [web/report/services/index.html](#)
- [web/report/store/index.html](#)
- [web/report/store/slices/index.html](#)
- [web/report/styles/index.html](#)

Rerun commands

```
dotnet test backend/tests/UnitTests/FamilyVeda.UnitTests.csproj --no-restore --collect:"XPlat Code Coverage" --results-directory docs/QMSE/evidence/2026-10-08-coverage/backend-unit
dotnet test backend/tests/IntegrationTests/FamilyVeda.IntegrationTests.csproj --no-restore --collect:"XPlat Code Coverage" --results-directory docs/QMSE/evidence/2026-10-08-coverage/backend-integration
cd web
npx vitest run --coverage --maxWorkers=2 --coverage.reportsDirectory=../docs/QMSE/evidence/2026-10-08-coverage/web/report
cd ../mobile
flutter test --no-pub --coverage --coverage-path=../docs/QMSE/evidence/2026-10-08-coverage/mobile/lcov.info
```

Local performance run

ApacheBench is selected for authenticated, read-only API baselines: it produces reproducible concurrency, throughput, response-time and failure measurements. Assignment PDF pp.1–2 lists examples and “other appropriate monitoring/testing tools”; it does not exclusively require k6/JMeter. This run does not measure write paths, hosted-agent latency, sustained/ramped traffic or production SLAs.

Fresh disposable PostgreSQL, local API port5060, 14 endpoints x1,000 requests, concurrency25. All14,000 requests completed with **zero failures and zero non-2xx**. p95 range **1–220ms**; slowest p95 is Family triage cases. No blanket performance acceptance was inferred.

[Full endpoint metrics](#).

```
# Password supplied privately; never put its value in the report.
FV_TEST_API_BASE=http://127.0.0.1:5060/api/v1 FV_LOAD_REQUESTS=1000 FV_LOAD_CONCURRENCY=25 python3 scripts/e2e/local_load_profile.py
```

Current ZAP scan

Independent disposable database/API5061, OpenAPI scan with synthetic Head authentication. **117 PASS / 0 FAIL / 2 WARN**, runner exit2 for warnings. High/Medium alert types:0; Low:2 types/4 instances; Informational:3 types. Retained low findings: CORP missing/invalid and Swagger unexpected Content-Type. No code/security changes were made for this scan.

184 imported URLs (456 URLs in total per the scan console); unresolved resource placeholders produced480 client-error instances. Generic schema-name import warnings and Head-only authentication limit coverage. Doctor/Admin scans are not claimed.

- [Scan summary](#)
- [Token-free reproduction instructions](#)
- [ZAP Markdown report](#)
- [ZAP HTML report](#)
- [ZAP JSON report](#)
- [Scan console](#)

Environment observation

Chrome rejects API port5060 with ERR_UNSAFE_PORT. Direct API/performance requests on5060 worked; connected browser E2E uses5080 and React5175 instead. This is a test-environment setup correction, not an application defect fix.

Doctor and Admin ZAP scans — 8 October

Same method as the Head scan, each on a fresh disposable database and local API (port 5091), authenticated with the seeded synthetic Doctor and Admin accounts.

Role	PASS	FAIL	WARN	High	Medium	Low	Informational
Head	117	0	2	0	0	2 types / 4 instances	3 types
Doctor	117	0	2	0	0	2 types / 7 instances	3 types (459 client-error responses)
Admin	117	0	2	0	0	2 types / 7 instances	3 types (469 client-error responses)

Role-only routes answered 200 under the matching token (/doctors/me, /doctors/me/cases; /admin/doctors, /admin/family-heads). The two Low warning types (Cross-Origin-Resource-Policy header, unexpected Content-Type on Swagger) are the same for all roles and remain open. Evidence: [Doctor](#) · [Admin](#).

Write-path and ramped load — 8 October

Script scripts/e2e/local_write_load_profile.py, local API and disposable database, synthetic Family Head.

Write path	Requests	Result	p50	p95
Create symptom episode	20	20 × 201	3.8 ms	6.6 ms
Submit triage case	20	20 × 202	10.9 ms	16.4 ms

Ramp on GET /dashboard/family, 2,000 requests per step, zero failed and zero non-2xx at every step:

Concurrency	Requests/s	p50	p95	p99
10	371	24 ms	45 ms	70 ms
25	432	53 ms	90 ms	113 ms
50	447	104 ms	165 ms	227 ms
100	437	208 ms	367 ms	407 ms

Interpretation: throughput levels off at about 430–450 requests/s from concurrency 25, after which extra load only adds latency. At 100 concurrent connections a separate `psql` session was refused with "too many clients", so the database connection limit is the first resource to saturate; the API itself returned no errors. No LLM keys were supplied in this run, so the 20 submitted cases ended `FailedSafe / AGENT_UNAVAILABLE` and agent latency is not part of these figures. Limits: one local machine, Debug build with SQL logging, one read endpoint, sequential writes. Evidence: [summary and rerun command](#) · [raw output](#).

Connected client E2E execution — 8 October 2026

Scope and outcome

Real Flutter iOS simulator and React Chrome clients used the same fresh local API5080/PostgreSQL database. UI automation used Computer Use, supplemented by authenticated HTTP assertions and actual persisted case identifiers. Synthetic accounts/data only; no production records/decisions changed. This is an executed tool-based integration journey, not a passing automated hosted-LLM certification.

Overall: executed, Failed/Partial. Fresh hosted submission failed safely; seeded approval propagated, but revealed internal-note exposure and stale-delay UI defects. Do not mark the whole workflow Pass.

Scenario	Actual outcome	Status
Flutter submit fresh synthetic case0021 → web doctor queue	Same persisted ID visible in both clients; guidance HTTP404	Pass for transport and pre-approval gate
Hosted processing case0021	FailedSafe / AGENT_UNAVAILABLE; Gemini failed, fallback config unavailable; guidance still404	Safe-failure invariant Pass; normal hosted completion Failed
Seeded case0003 before decision	Patient guidance404; same case visible in Flutter and web doctor desk	Pass
Web approve seeded case0003 → backend → Flutter	Web saved decision; backend statusApproved, selected final guidance endpoint200; Flutter showed response received	Transport/decision propagation Pass
Doctor-only note confidentiality	Exact internal note returned in patient DTO and rendered on Flutter patient progress	Failed, High defect
Approved guidance navigation	Old DOCTOR_RESPONSE_DELAY persisted; Flutter still showed delayed-review warning and omitted approved guidance button	Failed, High functional defect

Correlated identifiers

- Fresh0021: b76cf4c5-aac1-414d-b231-a1bf59aaa1bc — submitted in this session from Flutter.
- Seeded0003: c5d100ec-860d-4130-baf4-6ac9d279a30a — pre-existing seed; not submitted or processed by a hosted model in this session. Seed and fresh cases are not combined into one passing happy path.

Reproduce

1. Create an isolated PostgreSQL database and apply existing migrations/seeds; use private environment credentials.
2. Start local API on5080, React with VITE_API_BASE_URL=http://127.0.0.1:5080/api/v1 on5175; use localhost browser origin configured in CORS.
3. Run Flutter iOS with --dart-define=API_BASE_URL=http://127.0.0.1:5080/api/v1.
4. Login as synthetic Adult in Flutter, submit the documented mild synthetic fixture. Correlate displayed case number with API/DB ID; assert approved-guidance404 before approval.

5. Observe same case in web Doctor queue. Hosted availability may differ on rerun; record actual outcome. Never bypass FailedSafe to force approval.
6. For independent seeded decision propagation, open Adult seeded pending case0003, assert404, open same case in web Doctor Approval Desk. Select existing in-person-review template; put harmless unique marker in Internal Clinical Notes. Approve and confirm in disposable environment.
7. Refresh Flutter via Dashboard → Triage → same case. Compare patient DTO and approved-guidance endpoint. Internal note must be absent from patient response; approved button must be present. Current run failed both expectations.

Assertion and response evidence

- [Fresh preapproval assertion](#)
- [Fresh safe failure and seeded pending denial](#)
- [Patient postapproval responses](#)
- [Postapproval privacy/navigation assertions](#)
- [Patient accessibility evidence](#)

Supplemental API journey

Existing `scripts/e2e/synthetic_portal_journey.py` was executed on the same local target and failed at its expected404 assertion: the Head received HTTP200 with an empty adult-record list. This response did not expose the private record. Treat as a test expectation/API contract mismatch requiring investigation, not proof of successful private-record access or a completed journey. [Actual log](#). No assertion was weakened to manufacture a pass.

Light-mode screenshots

01-ios-local-adult-dashboard



iPhone 18 Pro Max
iOS 27.0



2:43



 **Dashboard**

 My Family

 Health Records



Family Veda

AP



WORKSPACE OVERVIEW

● ACTIVE



Amaya Perera

AdultMember

Active profile • Tap for details



Describe a symptom



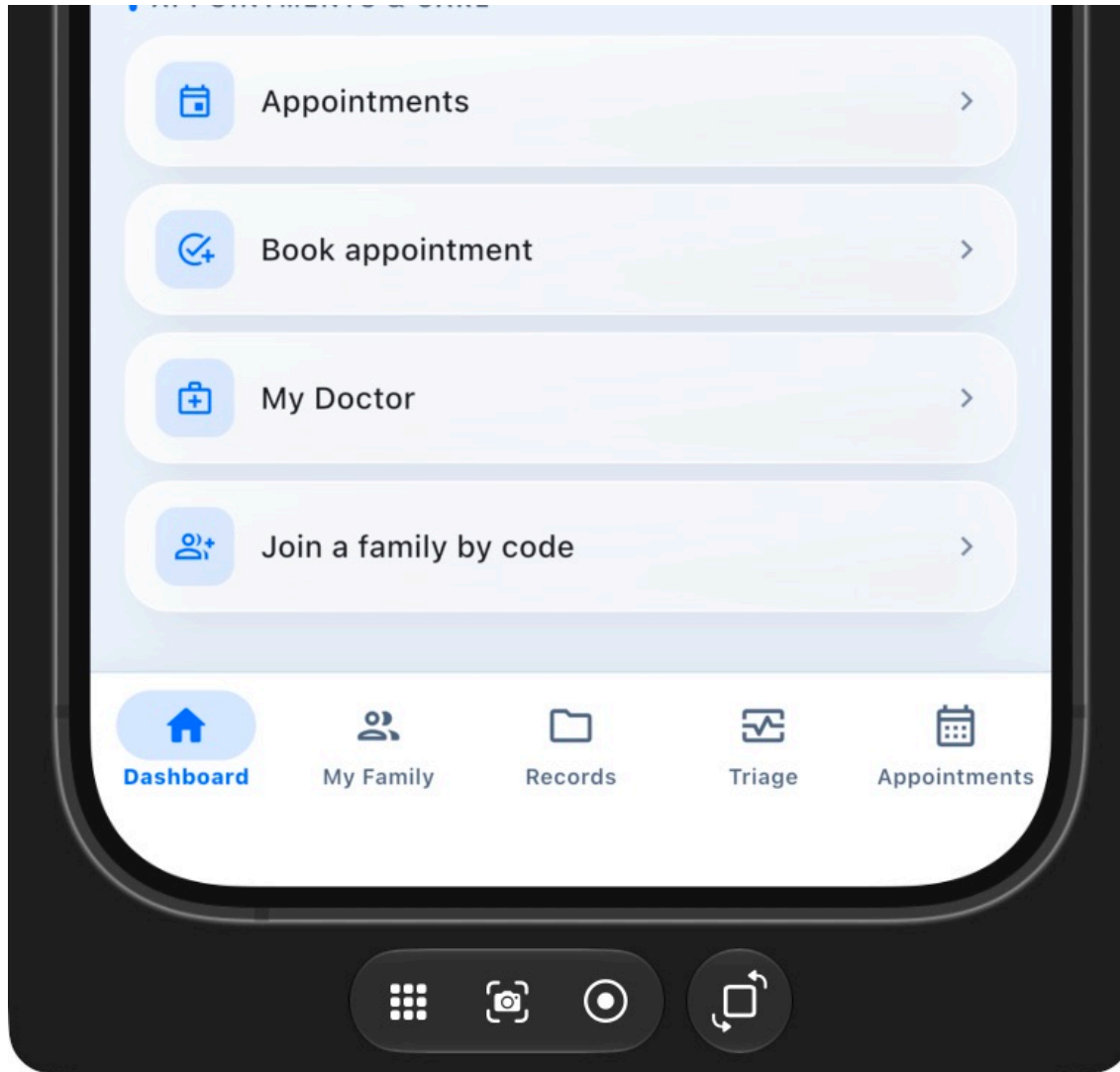
A verified doctor reviews every response before it reaches you.

Next appointment: Confirmed

Open cases: 1 • Approved guidance: 1

Family doctor: Dr. Dr. Synthetic Perera

APPOINTMENTS & CARE



02-ios-review-before-submit



iPhone 18 Pro Max
iOS 27.0



2:44



< New symptom request



Symptoms



Details

3

Review

Check your request before submitting

Family member

Amaya Perera

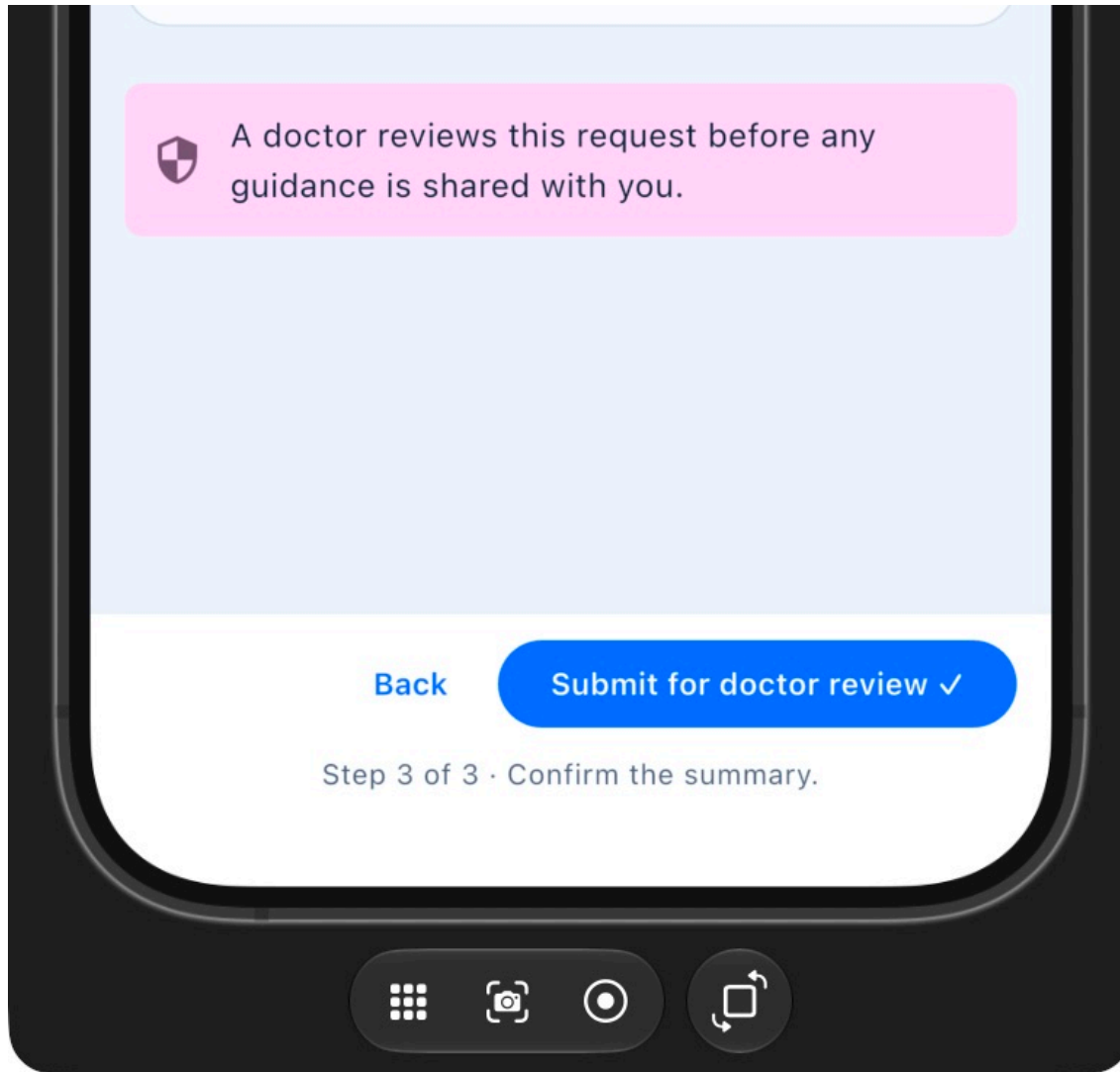
Symptoms and description

Feeling tired

Synthetic QMSE E2E 20261008: feeling tired for one day. Test fixture only.

Further details

Duration: 1 day · Severity: Mild



03-ios-submitted-awaiting-review



iPhone 18 Pro Max
iOS 27.0



2:44



< New symptom request

< Case progress

Request progress

Step 2 of 4

Submitted Oct 8, 2026



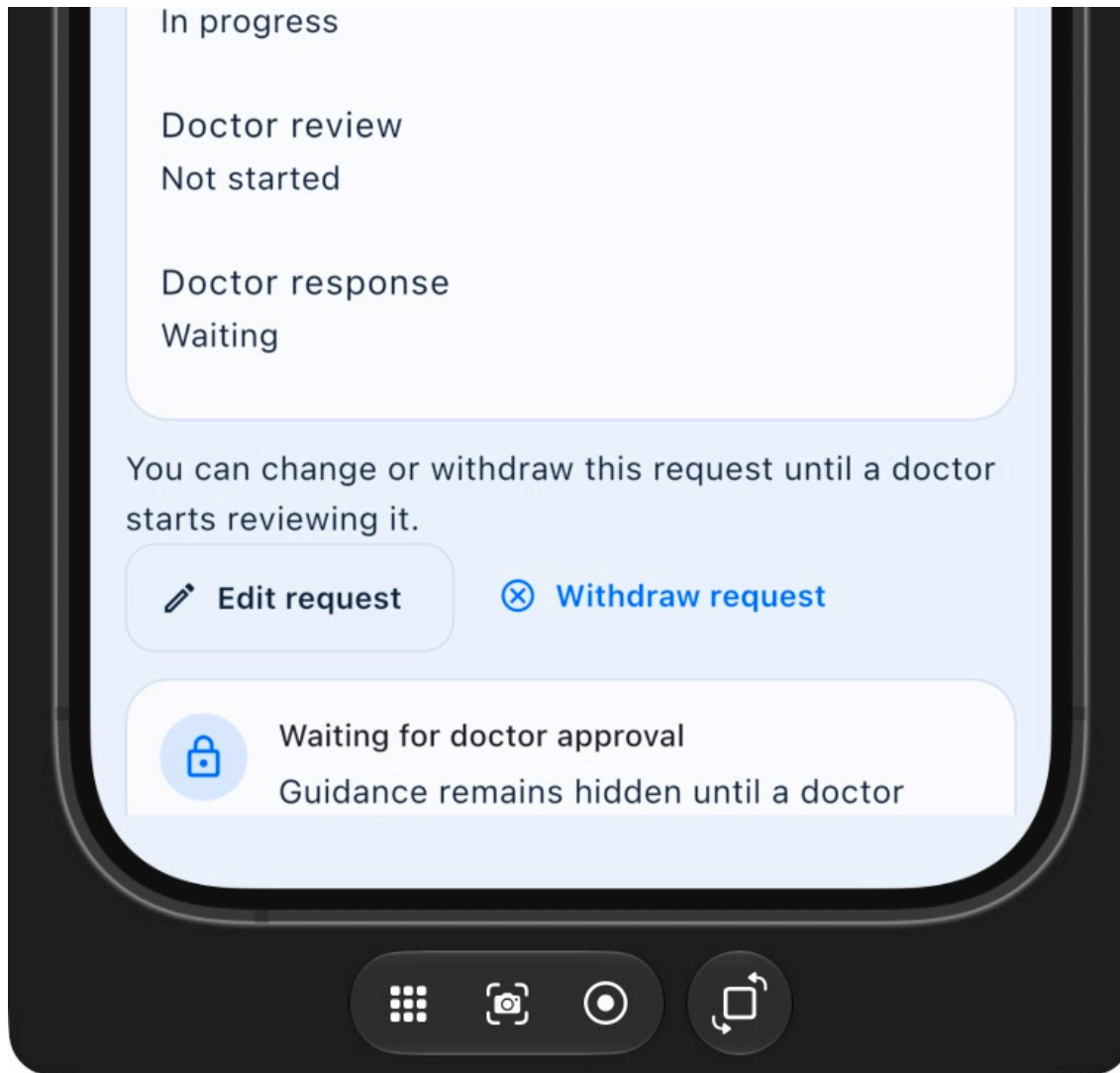
Request received

Oct 8, 2026 2:44 PM

Doctor received request

Oct 8, 2026 2:44 PM

AI processing



04-ios-case0021-no-guidance



iPhone 18 Pro Max
iOS 27.0



2:45



My Family

Health Records

Symptoms & Triage

Your requests

Track case progress and open approved guidance.

4 requests

✓ All

In review

Guidance ready

AP

Amaya Perera

Case 0021 · Oct 8, 2026



Being reviewed

AP

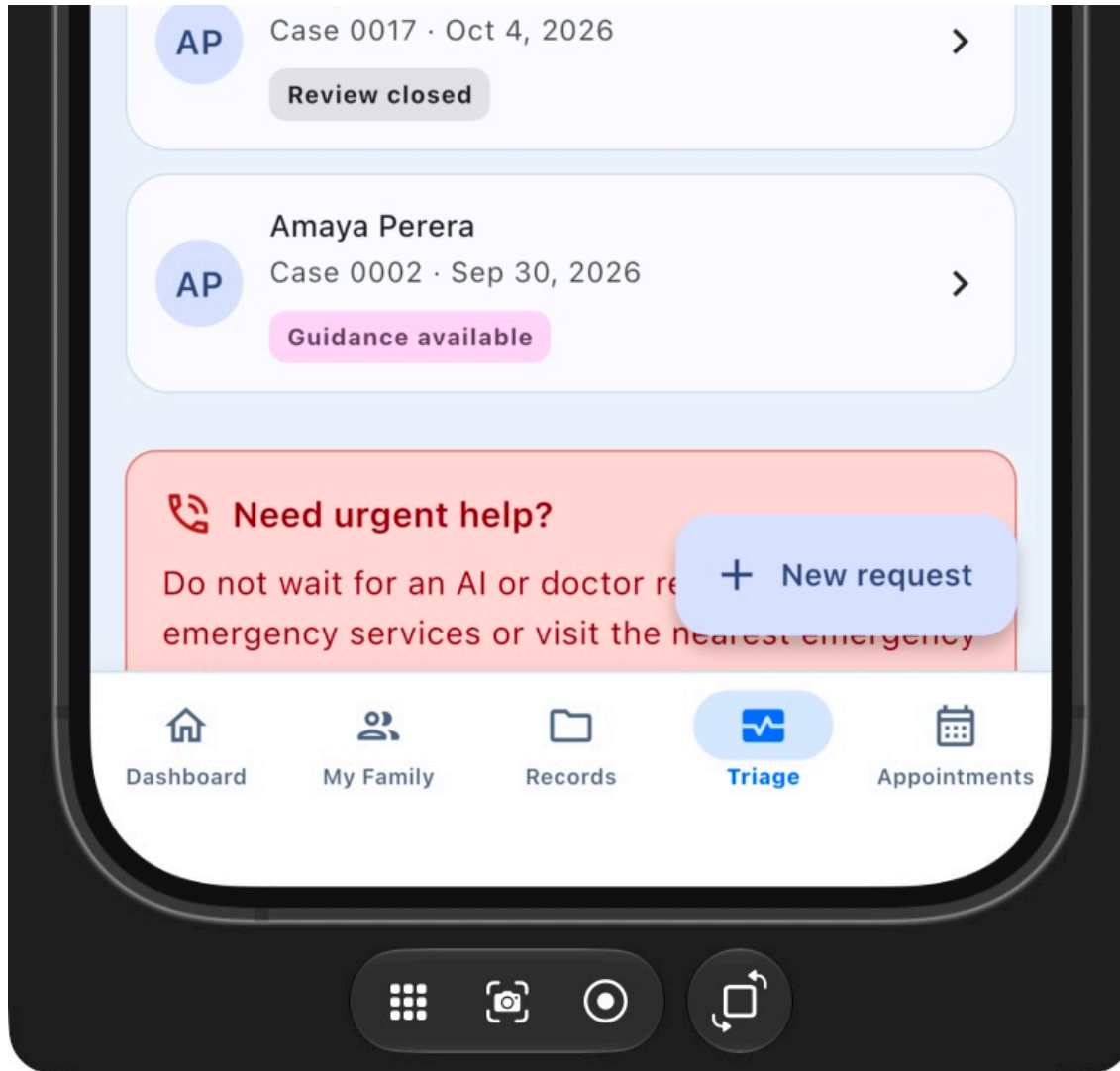
Amaya Perera

Case 0003 · Oct 7, 2026



Waiting for doctor review

Amaya Perera



05-web-doctor-case0021-processing



Triage Case Management

Manage incoming requests and cases assigned to you.

0021



Refresh

01

Processing Requests

02

Available Cases

06

My Active Cases

05

Awaiting Review

01

Emergency Referrals

All 15

Processing 1

Available 2

My Cases 6

Completed 5

Emergency 1

Priority

All priorities

Status

All statuses

Sort

Oldest first

Case 0021

Submitted Oct 8, 2:44 PM

AI processing

AI processing. Clinical review will become available after checks complete.

Selected case

Choose a case from the list to see its details and the next step.

06-web-case0021-safe-failure

Family Veda
DOCTOR PORTAL

DAY MODE

DR.

Dashboard

Calendar

My Families

Triage Cases

Approvals

Profile & Availability

Triage Case Management

Manage incoming requests and cases assigned to you.

0021

Refresh

01
Processing Requests

02
Available Cases

06
My Active Cases

05
Awaiting Review

01
Emergency Referrals

All 15

Processing 1

Available 2

My Cases 6

Completed 5

Emergency 1

Priority

Status

Sort

All priorities

All statuses

Oldest first

Case 0021

Submitted Oct 8, 2:44 PM

Processing stopped safely

Processing stopped safely. No AI guidance was released; in-person clinical review is needed.

Selected case

Choose a case from the list to see its details and the next step.

07-ios-case0021-safe-failure



iPhone 18 Pro Max
iOS 27.0



2:46



My Family

Health Records

~ Symptoms & Triage

Your requests

Track case progress and open approved guidance.

4 requests

✓ All

In review

Guidance ready

Amaya Perera

AP

Case 0021 · Oct 8, 2026



In-person care needed

Amaya Perera

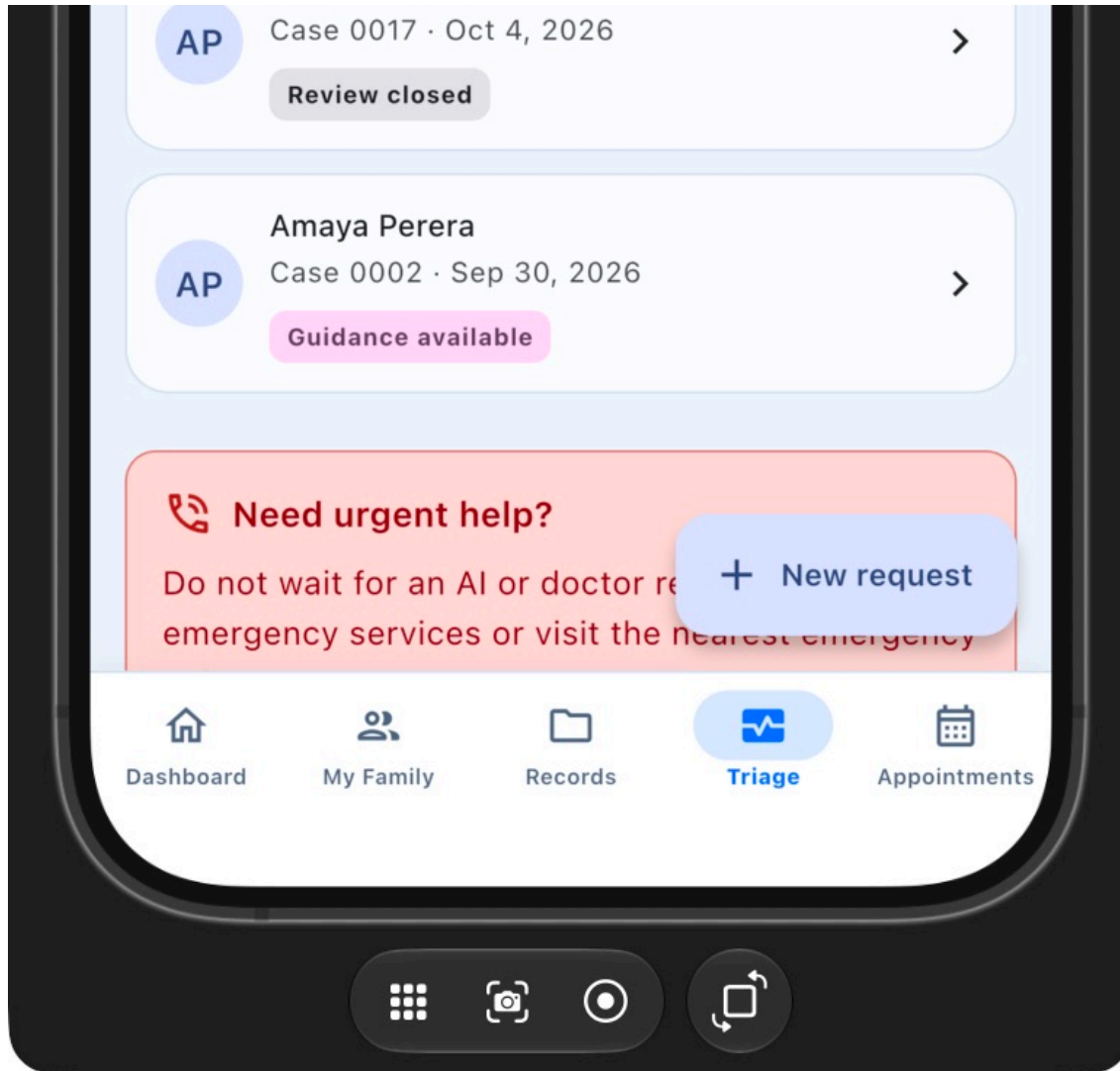
AP

Case 0003 · Oct 7, 2026



Waiting for doctor review

Amaya Perera



08-ios-seeded-case0003-before-approval



iPhone 18 Pro Max
iOS 27.0



2:47



 My Family

 Health Records

 Symptoms & Triage

Your requests

< Case progress

Request progress

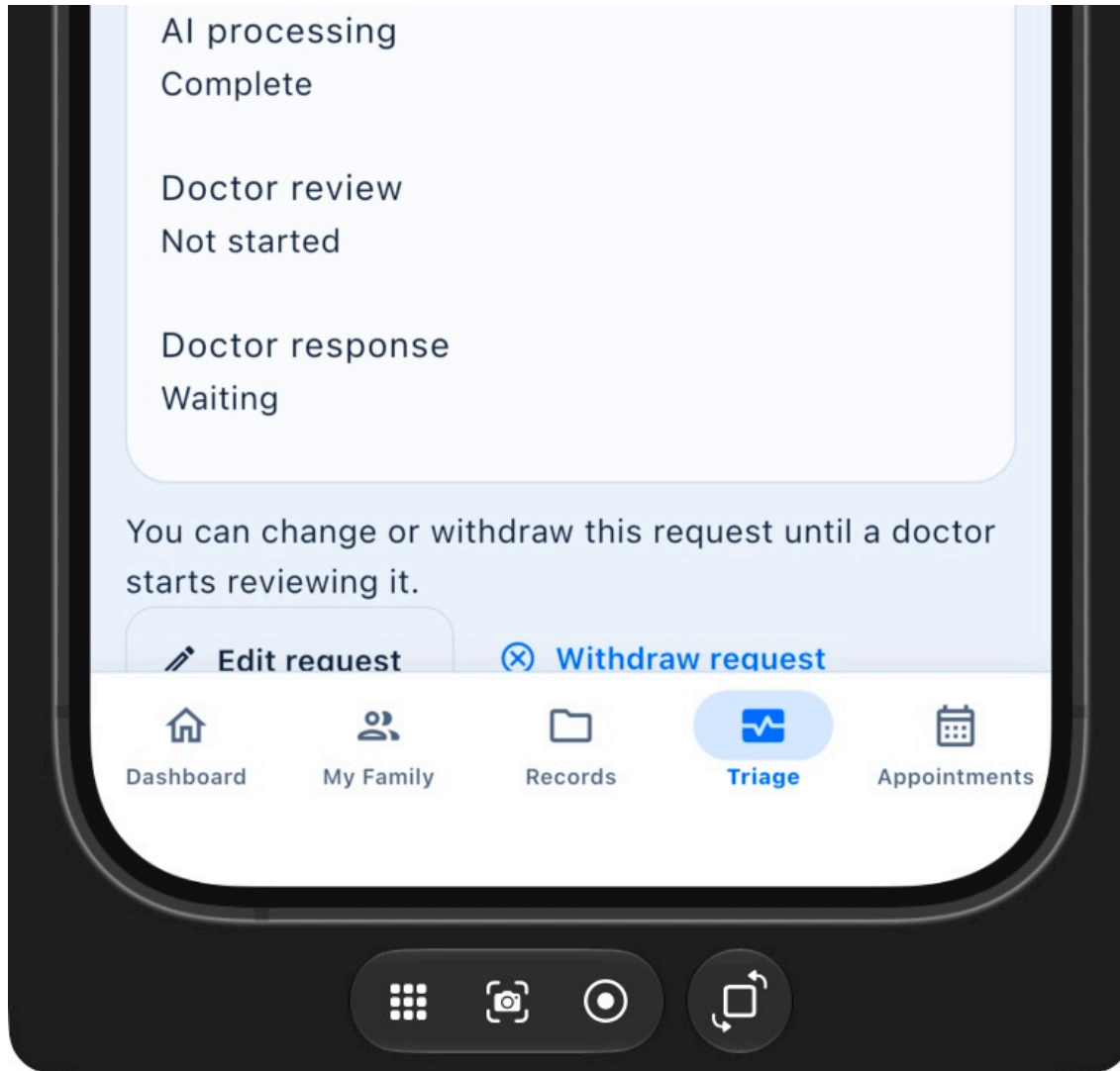
Step 2 of 4

Submitted Oct 7, 2026



Request received
Oct 7, 2026 6:33 PM

Doctor received request
Awaiting receipt



09-web-case0003-doctor-review



01
Processing Requests

02
Available Cases

06
My Active Cases

05
Awaiting Review

01
Emergency Referrals

All 15

Processing 1

Available 2

My Cases 6

Completed 5

Emergency 1

Priority

All priorities



Status

All statuses



Sort

Oldest first



Amaya Perera

Perera Family - Case 0003 - Submitted Oct 7, 6:33 PM

Routine

Awaiting clinical review

Open the case to see the submitted complaint.

Open Case

SELECTED CASE

Amaya Perera

Perera Family - Case 0003

Submitted Oct 7, 6:33 PM

Routine priority

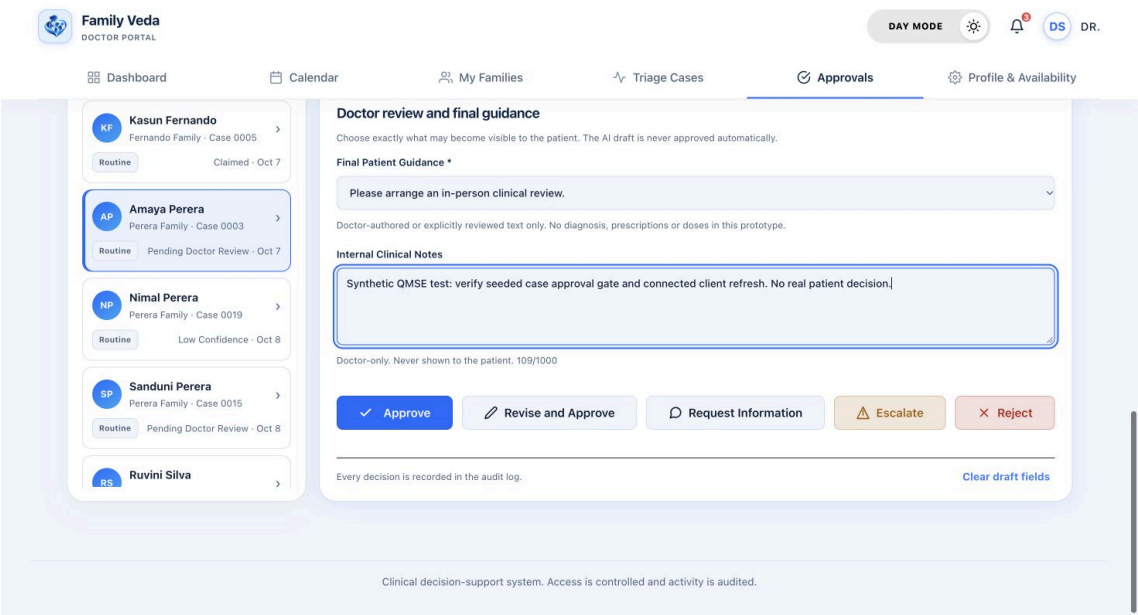
Awaiting clinical review

Patient context

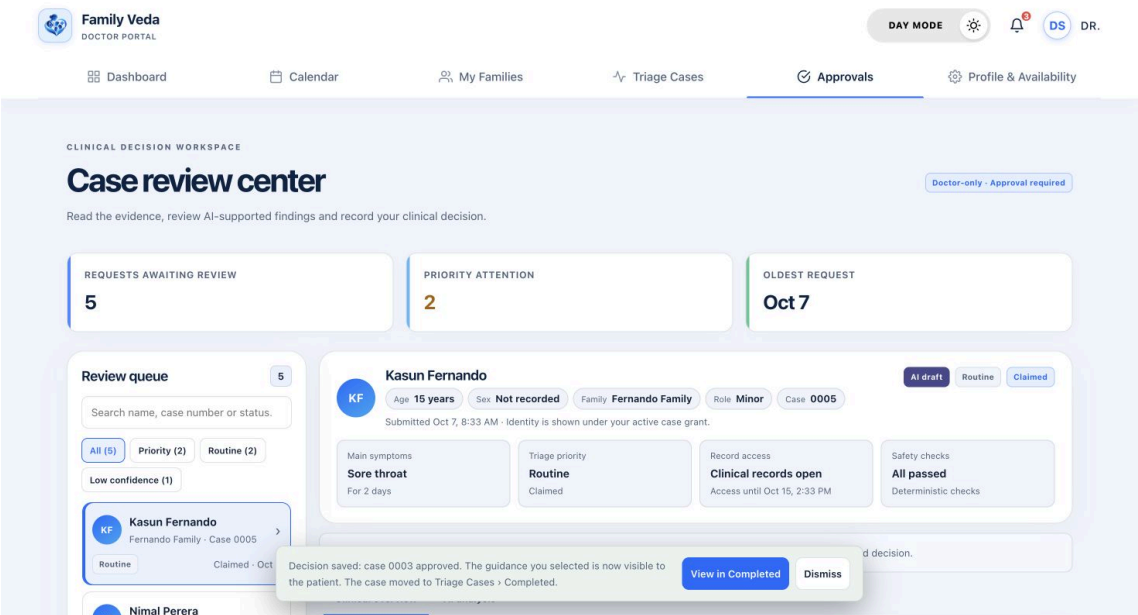
Patient request. Identity is limited to what your case grant allows.

Complaint

10-web-seeded-case0003-before-decision



11-web-seeded-case0003-approved



12-ios-case0003-approved-internal-notes-visible



iPhone 18 Pro Max
iOS 27.0



2:48



My Family

Health Records

~ Symptoms & Triage

Your requests

< Case progress

Request progress

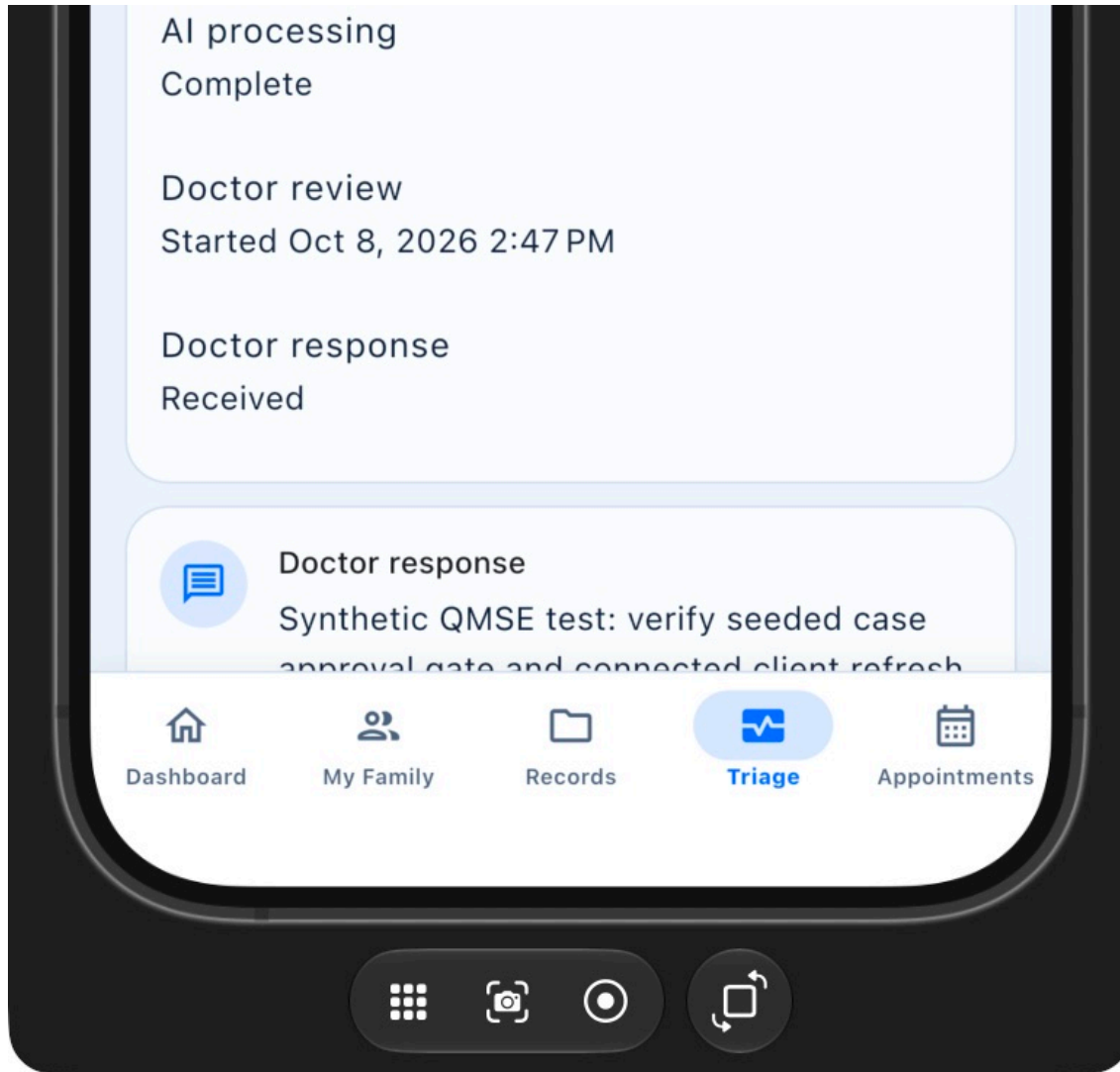
Step 4 of 4

Submitted Oct 7, 2026



Request received
Oct 7, 2026 6:33 PM

Doctor received request
Awaiting receipt



Hosted-agent golden journey — 8 October (after the defect fix)

The fresh case in the run above failed safe because the local API had no working provider configuration. The journey was repeated with the provider keys loaded, using `scripts/e2e/hosted_agent_golden_journey.py` against a disposable database and a local API (port 5095). The agents called the hosted Gemini model; all data is synthetic.

Variant	Case ID	Pipeline outcome	Checks
Seeded member with recorded vitals and labs	942a0b6d-3293-4d3b-8cfe-6e85b3ce3ef	PendingDoctorReview in 30.1 s; Context 0.90 and Analysis 0.88 confidence; draft advisory produced	11 passed, 0 failed

Variant	Case ID	Pipeline outcome	Checks
Newly registered member with no health data	a36a0387-89ce-4444-8257-bf437e0f3b02	LowConfidence in 20.2 s; agents report NoData; no draft advisory	10 passed, 0 failed

Checks in each run: guidance returns 404 before any decision; a family user gets 403 on the doctor-only review; every completed agent step is schema-valid and uses only allow-listed tools (none denied); guidance is still 404 after the agents finish; the family progress view carries step metadata only; after the doctor approves, the family reads the approved guidance; the internal doctor note is absent from every patient response; no overdue-review marker remains.

This is an API-driven run (no screens), so it complements the visual client run above rather than replacing it. Rerun: start a local API with the provider keys and `Seed__Enabled=true`, then `FV_TEST_PASSWORD=<private> FV_TEST_SEEDED_HEAD_EMAIL=demo-head@example.invalid python3 scripts/e2e/hosted_agent_golden_journey.py` (omit the e-mail variable for the new-member variant). Evidence: [seeded member](#) · [new member](#).

Defect log and retest evidence

Reconciled on 8 October 2026 against retained repository evidence. No tests were rerun during this document review. Results dated 5 October are historical and do not certify the latest working tree.

8. Defect Report and Retest Evidence

Table 8 is the defect report: each defect recorded during execution of the test plan, with its reproduction steps, cause, fix, retest outcome and current status.

Table 8 – Defect and Retest Log

ID	Description	Severity / priority	Steps to reproduce	Root cause	Fix and retest result	Status	Evidence
D-001	Doctor acceptance returned HTTP 500	High / P1	Run the synthetic API journey to the doctor-acceptance step with a doctor already assigned	Duplicate assignment inserted for an already assigned doctor	Fixed; integration test returns HTTP 200 with one assignment and request Accepted	Closed	AuthAndPatientFlowTests (TC A2-API-02)
D-002	Flutter verification blocked	Blocker / P1	Run flutter pub get on Flutter 3.41.2 / Dart 3.11	camera ^0.12.1 requires Dart 3.12	SDK upgraded to Flutter 3.47.5 / Dart 3.13.4; analyze clean and tests pass	Closed	../evidence/2026-09-28/flutter-tests.txt
D-003	Doctor dashboard returned HTTP 500	High / P1	Confirm an appointment, then GET /api/v1/dashboard/doctor	Non-UTC DateTimeOffset day boundary passed to Npgsql	Query boundary changed to UTC; integration check and repeat journey passed	Closed	../evidence/2026-09-28/backend-integration.txt
D-004	Android appointment time shown in UTC	Medium / P2	On a Sri Lanka-time emulator, book 10:00 AM and open the appointment list; 4:30 AM is shown	Model displayed the API instant without converting to device-local time	Timestamp parsed to local time; unit test and emulator retest show 10:00 AM	Closed	docs/evidence/android_appointment_local_time.png
D-005	Golden-case test returned HTTP 409	Test design error / P3	Run the golden-case test, which called /claim on a case the primary doctor already held	Test used the shared-pool route for an already granted case	Test corrected to expect 409 and approve through the existing grant; 2/2 passed	Closed	GoldenCaseFlowTests
D-006	Hosted doctor dashboard rendered blank	High / P1	Sign in as the synthetic verified doctor on the	API returns an integer count; the client expected an array	Client contract corrected (PR #49) and redeployed; panel renders with no console errors; regression test added	Closed	../evidence/2026-09-28/web-tests.txt

ID	Description	Severity / priority	Steps to reproduce	Root cause	Fix and retest result	Status	Evidence
			hosted web app and open /dashboard				
D-007	Sample metrics shown above live doctor metrics	Medium / P2	After the D-006 fix, open the doctor /dashboard; hard-coded counts disagree with live counts	Static sample content rendered alongside the live panel	Route changed to render the live panel only (DashboardPage.tsx returns DoctorDashboardPanel for doctors). Retested on the current develop build against a local API on 2026-10-05 (final report, Section 4.3): live panel only. A sign-in retest on the hosted site was not performed	Closed (local retest)	final report Figure 4.8; docs/university/RELEASE_EVIDENCE_2026-09-28.md
D-008	Two vitals-panel web tests fail	Medium / P2	In web/, run npx vite --run src/pages/records/VitalsPanel.test.tsx at 613bf7e	The redesign of the vitals panel removed the history table and its type filter, but two tests still asserted them; the product behaved as designed and the tests were obsolete	The history-table assertion was removed from one test and the filter test was deleted; recorded readings remain covered by the per-vital dialog test. Retest: web suite passed in full	Closed	evidence/2026-10-05/web-tests.txt, web-tests-retest.txt
D-009	Web lint and production build fail	High / P1	In web/, run npm run lint then npm run build at 613bf7e	Unused Link import left in RecordsPage.tsx (ESLint no-unused-vars, TypeScript TS6133)	Import removed. Retest: lint exit 0, build exit 0	Closed	evidence/2026-10-05/web-lint-build.txt, web-lint-build-retest.txt
D-010	Patient vital charts drew a hard-coded "normal" band	High / P1 (clinical safety)	In the web app open Health Records → Vitals at 613bf7e; each chart shows a green reference band and limits	Fixed reference limits were written into the web client and passed to the chart. One fixed range cannot fit every member (age, pregnancy, measurement context), and showing it classifies a reading for the patient without a doctor, contrary to safety rules 1, 2 and 4. Found by automated pull-request review	Limits removed from the client; the chart now draws recorded values and their average only. A guard test asserts that no reference range is drawn. Retest: web suite 308 / 308, lint and build exit 0	Closed	VitalsPanel.test.tsx; pull request #160
D-011	Server error on a NULL character in a query string	Medium / P2	GET /api/v1/doctor/s/directory?search=%00 against the API	The character reached PostgreSQL, which cannot store it in text; the failure surfaced as HTTP 500 instead of a client error. Found by the OWASP ZAP scan	RequestHardeningMiddleware rejects such a query string with HTTP 400 problem details. Retest: integration test passed; ZAP rescan no longer reports a server error	Closed	RequestHardeningTests; evidence/2026-10-05/security-zap-summary.txt
D-012	X-Content-Type-Options header missing from API responses	Low / P3	Inspect the response headers of any API route	The header was never set. Found by the OWASP ZAP scan	The middleware adds nosniff to every response. Retest: integration test passed; ZAP rescan no longer reports it	Closed	RequestHardeningTests; zap-retest/

Priority follows severity: P1 blocks a user journey or the build, P2 misleads without blocking, and P3 affects only the test suite. D-009 is rated high because a failing build blocks the CI quality gate and the web deployment.

Two of these defects, D-003 and D-006, are instructive. D-003 was found only because the integration tests ran against real PostgreSQL, where Npgsql rejects a non-UTC offset that an in-memory provider would have accepted. D-006 was a contract mismatch between client and API that component tests with mocked data did not detect, and it was found by a live synthetic sign-in; a regression test now guards it. Related deployment findings (a hosted Swagger 500 from a duplicate schema identifier, and a notifications page that previously failed to load until a follow-up Neon migration) were also corrected and are recorded in the release evidence.

Current capture observations awaiting investigation

ID	Observation	Reproduction	Status / evidence
OBS-IOS-01	Adult Join Requests and Invitations showed a safe error	Synthetic Adult account → My Family → respective tab on 8 October	Open observation, not a confirmed root cause or fix; see gallery .
OBS-IOS-02	Simulator startup/freeze recovered after runtime relaunch	Earlier iOS capture session under heavy host load	Recovered runtime only; permanent app fix not established

Supplied defect claims needing corroboration

The supplied DEF-001–DEF-008 identifiers are separate from retained D-001–D-012 and must not replace them. Supplied claims about validator DI, Npgsql migration probing, Windows file modes, shell configuration, emergency prechecks, consent cache invalidation, Docker wrappers and large-PDF uploads require first-run logs, exact fix commit and retest evidence. DEF-008 has no detailed report in the supplied file. A stopped Docker daemon is an environment blocker; an in-memory fallback does not prove PostgreSQL integration. No new clinical/security defect or resolution is inferred from draft prose.

Newly observed connected-client defects — 8 October

QMSE-20261008-01 — Doctor-only internal note visible to patient

Severity: **HIGH**. Status: Fixed in commit `bed063f` ([PR #176](#)); automated retest Passed; connected-client visual retest not yet repeated.

Precondition: seeded Adult case0003 pending review in isolated DB. Web Doctor enters a harmless marker in Internal Clinical Notes (labelled doctor-only), approves with existing final-guidance template. Patient GET case returns the marker in `latestDecisionReason`; Flutter renders it under Doctor response. Expected: internal notes never enter patient DTO/UI. Actual: exact marker exposed. Source: `Infrastructure/Component3_Triage/Triage/TriageService.cs:353` maps `DoctorNotes` to `LatestDecisionReason`; `mobile/lib/screens/triage/case_status_screen.dart:98` and `web/src/pages/triage/TriagePage.tsx:592` render it. Web patient rendering is source-confirmed, not independently UI-run in this session.

QMSE-20261008-02 — Approved case retains delayed-review state and hides guidance button

Severity: **HIGH** functional. Status: Fixed in commit `bed063f` ([PR #176](#)); automated retest Passed; connected-client visual retest not yet repeated.

Same case became Approved; approved-guidance returned200 with the selected final text. `failureCode` remained `DOCTOR_RESPONSE_DELAY`. Flutter delayed branch displayed doctor-not-responded warning and suppressed View approved guidance. Expected: approval clears/reconciles obsolete delay and offers approved guidance. Actual: contradictory terminal/progress state. See `mobile/lib/screens/triage/case_status_screen.dart:83–115` and current patient response/accessibility evidence.

[Connected E2E evidence, steps and assertions](#). The observations were recorded before any fix. Personal discoverer, student fixer and retester fields remain blank; these observations were collected by Codex.

Fix and retest evidence — 8 October

Step	Tool	Actual result	Evidence
Before fix	xUnit + Testcontainers PostgreSQL, ApprovedCase_NeverReturnsInternalDoctorNotes_AndClearsStaleDelayMarker	Failed: patient response contained the marker note in latestDecisionReason and failureCode DOCTOR_RESPONSE_DELAY	01-before-fix-integration.txt
After fix, API	Full integration suite	31 passed, 0 failed (30 earlier tests plus the new regression test)	02-after-fix-integration.txt
After fix, React	Vitest + React Testing Library, TriagePage.test.tsx	12 passed, including "never renders internal doctor notes in the patient doctor-response card"	03-after-fix-web.txt
After fix, Flutter	flutter_test, case_status_screen_test.dart	4 passed, including "approved case hides a stale delay marker and never shows internal doctor notes"	04-after-fix-flutter.txt

Fix summary: the patient case DTO no longer carries Approval.DoctorNotes; both clients stopped rendering latestDecisionReason; a final doctor decision clears DOCTOR_RESPONSE_DELAY, and the Flutter status screen ignores that marker once a decision exists. No migration. Consequence to note: patients now see only the fixed decision label for Request-information and Reject decisions, not free text.

Rerun: dotnet test backend/tests/IntegrationTests/FamilyVeda.IntegrationTests.csproj --filter "FullyQualifiedName~ApprovedCase_NeverReturnsInternalDoctorNotes".

QMSE-20261008-03 — Adult sees errors on Head-only family tabs (was OBS-IOS-01)

Severity: **MEDIUM**. Status: Fixed; automated retest Passed; not re-run on a device.

Cause: the Flutter Members screen showed Join Requests and Invitations to every user and loaded two providers whose endpoints are Head-only, so an Adult got "Something went wrong". Fix: those tabs and providers are used only for a Family Head; Adults keep Members, incoming invitations and Settings. The React app already hid these for Adults.

Step	Result	Evidence
Before fix, new widget test "hides Head-only tabs and never calls Head-only providers for an Adult"	Failed	05-before-fix-flutter-adult-family.txt
After fix, members_screen_test.dart	2 passed	06-after-fix-flutter-adult-family.txt
Full Flutter suite and analyze	258 passed; no issues	same run

QMSE-20261008-04 — API journey script expected 404 for a private record list

Severity: **LOW** (test defect, not a product defect). Status: Fixed.

scripts/e2e/synthetic_portal_journey.py expected 404 when a Head lists an adult's records. The tested contract (AdultReportSharingTests) is an empty list with 200 for lists and 404 for single-record reads; no private data was returned. The script now asserts the empty list, and its doctor lookup no longer depends on a stale display name. The script has not been re-run end to end since this change.

Individual test contributions — repository evidence

Compiled 8 October 2026. Git author/file history is evidence of recorded changes, not proof of who personally ran tests, independent authorship, viva understanding or ownership of every changed component. S1–S4 handle mapping comes from docs/OWNERSHIP.tsv. Scope tags alone do not establish authorship.

Repository: <https://github.com/sahansbandara/Family-Veda-SEF-Project>

Current tool execution

Codex executed xUnit, Testcontainers/PostgreSQL, Vitest/RTL and flutter_test on the current working tree. These runs cannot be claimed as personal student demonstrations. All four suites passed: 430 unit + 30 integration + 343 React + 256 Flutter = 1,059 tests. See [execution evidence](#).

Author-specific test changes

S1 — Samaranayaka S.G.V.S

Commit / date	Recorded change	Test files changed	Tool indicated by source
e678bf1 / 2026-09-23	test(s1): cover the family, identity and consent providers and screens	mobile/test/main_test.dart mobile/test/providers/active_member_provider_test.dart mobile/test/providers/auth_provider_test.dart mobile/test/router/route_guard_test.dart mobile/test/screens/login_screen_test.dart mobile/test/services/api_services_test.dart mobile/test/services/secure_storage_test.dart	flutter_test
eadc2e5 / 2026-09-23	test(s1): cover the family, identity and consent web screens	web/src/pages/family/OnboardingPage.test.tsx web/src/routes/AppRouter.test.tsx web/src/store/slices/authSlice.test.ts	Vitest/RTL

Student-confirmed field	Value
Personally demonstrated command/date	
Personal role in defect discovery/fix/retest	
Viva explanation and modifications demonstrated	
CLEAR declaration	

S2 — Fernando K.R.N / Ranidu Fernando

Commit / date	Recorded change	Test files changed	Tool indicated by source
43ca434 / 2026-09-22	test(s2): cover the health records and extraction providers and screens	mobile/test/screens/records_screen_test.dart	flutter_test
2731780 / 2026-09-22	test(s2): cover the health records and extraction web screens	web/src/pages/records/RecordsPage.test.tsx	Vitest/RTL
ef35ed8 / 2026-09-20	test(s2): cover extraction agent, record sorting, and records UI	mobile/test/screens/records_screen_test.dart web/src/pages/records/RecordsPage.test.tsx	Vitest/RTL, flutter_test

Student-confirmed field	Value
Personally demonstrated command/date	
Personal role in defect discovery/fix/retest	
Viva explanation and modifications demonstrated	
CLEAR declaration	

S3 — Karunathilaka K.D.J.C / Jani6969

Commit / date	Recorded change	Test files changed	Tool indicated by source
3e137e7 / 2026-09-23	test(s3): cover the triage and agent orchestration providers and screens	mobile/test/providers/patient_providers_test.dart mobile/test/screens/async_state_views_test.dart mobile/test/screens/case_status_screen_test.dart mobile/test/screens/home_screen_test.dart	flutter_test
d09383e / 2026-09-23	test(s3): cover the triage and agent orchestration web screens	web/src/components/shared/ViewState.test.tsx	Vitest/RTL

Student-confirmed field	Value
Personally demonstrated command/date	
Personal role in defect discovery/fix/retest	
Viva explanation and modifications demonstrated	
CLEAR declaration	

S4 — ImSahanS; handle @sahansbandara

Commit / date	Recorded change	Test files changed	Tool indicated by source
f002316 / 2026-10-04	test(s4): cover the name-first case title and anonymous pool fallback (AI co-authored)	mobile/test/models/doctor_queue_case_reference_test.dart	flutter_test
e877454 / 2026-10-04	test(s4): cover the doctor menu after a fresh sign-in through the real router (AI co-authored)	mobile/test/router/doctor_shell_after_login_test.dart	flutter_test

Commit / date	Recorded change	Test files changed	Tool indicated by source
fa6c357 / 2026-09-22	test(s4): cover the familial risk and clinical approval providers and screens	mobile/test/models/approved_guidance_test.dart mobile/test/providers/guidance_provider_test.dart mobile/test/screens/approved_guidance_screen_test.dart mobile/test/screens/emergency_screen_test.dart	flutter_test
977c822 / 2026-09-22	test(s4): cover the familial risk and clinical approval web screens	web/src/pages/doctor/ApprovalsPage.test.tsx	Vitest/RTL

Student-confirmed field	Value
Personally demonstrated command/date	
Personal role in defect discovery/fix/retest	
Viva explanation and modifications demonstrated	
CLEAR declaration	

Tool source inventory

I Tool I Existing source/configuration I Evidence I |---|---| I xUnit + Moq I backend/tests/UnitTests/FamilyVeda.UnitTests.csproj and component test folders I Current TRX + text log I I xUnit + PostgreSQL Testcontainers I backend/tests/IntegrationTests/FamilyVeda.IntegrationTests.csproj I Current TRX + text log; real Docker required I I Vitest + React Testing Library I web/package.json, web test sources I Current 57 files / 343 tests I I flutter_test I mobile/pubspec.yaml, mobile/test/ I Current 256 tests I I ApacheBench I scripts/e2e/local_load_profile.py I 14-endpoint output, rerun 8 October (evidence/2026-10-08-nonfunctional/performance-load-profile.txt) I I OWASP ZAP I Retained first-run/retest reports I Head-token API scan, 5 October first run/retest and 8 October rerun (evidence/2026-10-08-zap/) I

Defect/fix/retest traceability

D-008/D-009: first failing React tests/lint/build and successful retests retained in 5 October evidence. D-010: VitalsPanel guard test and PR #160. D-011/D-012: ZAP first run → [request hardening fix 7a90a77](#) → RequestHardeningTests → historical ZAP rescan and current 30-test integration pass. These are project traces; the student who discovered or retested each defect remains blank unless personally confirmed.

Computer Use verified the GitHub fix page; [commit screenshot](#). The GitHub develop page showed a newer merge than local HEAD; current tests certify the recorded local working tree, not that later remote merge.

AI attribution

The repository contains Claude-authored commits and AI co-author trailers. Preserve those identities. AI-authored changes must not be reassigned to a student just because the component tag is S1–S4. Students must disclose their actual use and understanding.

Complete test file inventory

- [backend/tests/IntegrationTests/Component1 Family/AuthAndPatientFlowTests.cs](#)
- [backend/tests/IntegrationTests/Component1 Family/DoctorRegistrationTests.cs](#)
- [backend/tests/IntegrationTests/Component1 Family/MigrationTests.cs](#)
- [backend/tests/IntegrationTests/Component1 Family/RegistrationFlowTests.cs](#)
- [backend/tests/IntegrationTests/Component2 Records/AdultReportSharingTests.cs](#)
- [backend/tests/IntegrationTests/RequestHardeningTests.cs](#)
- [backend/tests/IntegrationTests/Shared/GoldenCaseFlowTests.cs](#)
- [backend/tests/UnitTests/AgentOutputPromptTests.cs](#)
- [backend/tests/UnitTests/CloudflareClientTests.cs](#)
- [backend/tests/UnitTests/Component1 Family/ConsentStateMachineTests.cs](#)
- [backend/tests/UnitTests/Component1 Family/FamilyAccessTests.cs](#)
- [backend/tests/UnitTests/Component1 Family/FamilyHeadTransferServiceTests.cs](#)
- [backend/tests/UnitTests/Component1 Family/FamilyLifecycleServiceTests.cs](#)
- [backend/tests/UnitTests/Component1 Family/FamilyServicePrivacyTests.cs](#)
- [backend/tests/UnitTests/Component1 Family/IncomingInvitationServiceTests.cs](#)
- [backend/tests/UnitTests/Component1 Family/ProfileServiceTests.cs](#)
- [backend/tests/UnitTests/Component1 Family/RegistrationValidatorTests.cs](#)
- [backend/tests/UnitTests/Component1 Family/ToolDispatcherTests.cs](#)
- [backend/tests/UnitTests/Component1 Family/ToolRegistryTests.cs](#)
- [backend/tests/UnitTests/Component2 Records/GoogleDriveReportStorageTests.cs](#)
- [backend/tests/UnitTests/Component2 Records/LabExtractionParserTests.cs](#)
- [backend/tests/UnitTests/Component2 Records/LabExtractionSafetyTests.cs](#)
- [backend/tests/UnitTests/Component2 Records/LabExtractionWorkerTests.cs](#)
- [backend/tests/UnitTests/Component2 Records/LabReportDurableStorageTests.cs](#)
- [backend/tests/UnitTests/Component2 Records/LabReportTrashTests.cs](#)
- [backend/tests/UnitTests/Component2 Records/PdfReportLimitTests.cs](#)
- [backend/tests/UnitTests/Component2 Records/RecordServiceLabReviewTests.cs](#)
- [backend/tests/UnitTests/Component2 Records/VitalReferenceRangesTests.cs](#)
- [backend/tests/UnitTests/Component3 Triage/CaseSlaProcessorTests.cs](#)
- [backend/tests/UnitTests/Component3 Triage/ChatCompletionsLlmClientTests.cs](#)
- [backend/tests/UnitTests/Component3 Triage/NotificationServiceTests.cs](#)
- [backend/tests/UnitTests/Component3 Triage/TriageOrchestratorEmergencyTests.cs](#)
- [backend/tests/UnitTests/Component3 Triage/TriageOrchestratorNoDataTests.cs](#)
- [backend/tests/UnitTests/Component3 Triage/TriageOrchestratorSchemaTests.cs](#)
- [backend/tests/UnitTests/Component3 Triage/TriageReviewReasonsTests.cs](#)
- [backend/tests/UnitTests/Component3 Triage/TriageSubmissionLifecycleTests.cs](#)
- [backend/tests/UnitTests/Component3 Triage/TriageWorkerRecoveryTests.cs](#)

- [backend/tests/UnitTests/Component4 Clinical/AdultPrivacyAuthorizationTests.cs](#)
- [backend/tests/UnitTests/Component4 Clinical/AppointmentServiceTests.cs](#)
- [backend/tests/UnitTests/Component4 Clinical/CaseGrantPolicyTests.cs](#)
- [backend/tests/UnitTests/Component4 Clinical/ClinicalCasePoolPrivacyTests.cs](#)
- [backend/tests/UnitTests/Component4 Clinical/ClinicalEmergencyReferralTests.cs](#)
- [backend/tests/UnitTests/Component4 Clinical/ClinicalRuleTableTests.cs](#)
- [backend/tests/UnitTests/Component4 Clinical/DoctorWorkspaceServiceTests.cs](#)
- [backend/tests/UnitTests/Component4 Clinical/EmergencyFollowUpTests.cs](#)
- [backend/tests/UnitTests/Component4 Clinical/FamilialRiskAgentTests.cs](#)
- [backend/tests/UnitTests/Component4 Clinical/FamilialRiskPolicyTests.cs](#)
- [backend/tests/UnitTests/Component4 Clinical/FamilyDoctorServiceTests.cs](#)
- [backend/tests/UnitTests/Component4 Clinical/JoinRequestServiceTests.cs](#)
- [backend/tests/UnitTests/Component4 Clinical/LabRangeClassifierTests.cs](#)
- [backend/tests/UnitTests/Component4 Clinical/PortalDashboardAndNotificationTests.cs](#)
- [backend/tests/UnitTests/Component4 Clinical/PortalDashboardMockupFieldsTests.cs](#)
- [backend/tests/UnitTests/Component4 Clinical/SafetyValidationServiceTests.cs](#)
- [backend/tests/UnitTests/GeminiKeyPoolTests.cs](#)
- [backend/tests/UnitTests/LlmFallbackCancellationTests.cs](#)
- [backend/tests/UnitTests/LlmFallbackClientTests.cs](#)
- [backend/tests/UnitTests/ProviderRedirectTests.cs](#)
- [backend/tests/UnitTests/Shared/Phase1bSeedTests.cs](#)
- [backend/tests/UnitTests/Shared/VivaDemoSeederTests.cs](#)
- [mobile/test/main test.dart](#)
- [mobile/test/models/app notification test.dart](#)
- [mobile/test/models/appointment test.dart](#)
- [mobile/test/models/approved guidance test.dart](#)
- [mobile/test/models/doctor queue case reference test.dart](#)
- [mobile/test/models/family dashboard test.dart](#)
- [mobile/test/models/lab report test.dart](#)
- [mobile/test/models/record summary meta test.dart](#)
- [mobile/test/models/triage case test.dart](#)
- [mobile/test/models/vital test.dart](#)
- [mobile/test/providers/active member provider test.dart](#)
- [mobile/test/providers/auth provider test.dart](#)
- [mobile/test/providers/guidance provider test.dart](#)
- [mobile/test/providers/members provider test.dart](#)
- [mobile/test/providers/patient providers test.dart](#)

- [mobile/test/router/doctor shell after login test.dart](#)
- [mobile/test/router/route guard test.dart](#)
- [mobile/test/screens/appointments screen test.dart](#)
- [mobile/test/screens/approved guidance screen test.dart](#)
- [mobile/test/screens/async state views test.dart](#)
- [mobile/test/screens/book appointment member options test.dart](#)
- [mobile/test/screens/book appointment screen test.dart](#)
- [mobile/test/screens/care workflows test.dart](#)
- [mobile/test/screens/case status screen test.dart](#)
- [mobile/test/screens/cases screen test.dart](#)
- [mobile/test/screens/doctor calendar logic test.dart](#)
- [mobile/test/screens/doctor calendar screen test.dart](#)
- [mobile/test/screens/doctor families flow test.dart](#)
- [mobile/test/screens/doctor profile screen test.dart](#)
- [mobile/test/screens/doctor triage cases screen test.dart](#)
- [mobile/test/screens/emergency screen test.dart](#)
- [mobile/test/screens/family doctor availability test.dart](#)
- [mobile/test/screens/home screen test.dart](#)
- [mobile/test/screens/login screen test.dart](#)
- [mobile/test/screens/members screen test.dart](#)
- [mobile/test/screens/my doctor screen test.dart](#)
- [mobile/test/screens/privacy screen test.dart](#)
- [mobile/test/screens/profile screen test.dart](#)
- [mobile/test/screens/public information screen test.dart](#)
- [mobile/test/screens/records screen test.dart](#)
- [mobile/test/screens/register screen test.dart](#)
- [mobile/test/screens/report library workflow test.dart](#)
- [mobile/test/screens/report trash test.dart](#)
- [mobile/test/screens/request progress actions test.dart](#)
- [mobile/test/screens/vitals tab test.dart](#)
- [mobile/test/services/api services test.dart](#)
- [mobile/test/services/auth api error test.dart](#)
- [mobile/test/services/report reading poller test.dart](#)
- [mobile/test/services/secure storage test.dart](#)
- [mobile/test/widgets/app shell test.dart](#)
- [mobile/test/widgets/doctor/vital overview grid test.dart](#)
- [mobile/test/widgets/head dashboard section test.dart](#)

- [mobile/test/widgets/report_library_card_test.dart](#)
- [mobile/test/widgets/report_review_workspace_test.dart](#)
- [scripts/e2e/local_load_profile.py](#)
- [scripts/e2e/local_performance_check.py](#)
- [scripts/e2e/synthetic_portal_journey.py](#)
- [web/src/components/records/DeletedReports.test.tsx](#)
- [web/src/components/records/OriginalReportPreview.test.tsx](#)
- [web/src/components/records/PdfReportCanvas.test.tsx](#)
- [web/src/components/records/RecordedRangeVisual.test.tsx](#)
- [web/src/components/records/ReportLibrary.test.tsx](#)
- [web/src/components/records/ReportLibraryCard.test.tsx](#)
- [web/src/components/records/ReportReviewDialog.test.tsx](#)
- [web/src/components/records/ReportThumbnail.test.tsx](#)
- [web/src/components/shared/ViewState.test.tsx](#)
- [web/src/pages/admin/AdminDashboardPanel.test.tsx](#)
- [web/src/pages/auth/AuthPage.test.tsx](#)
- [web/src/pages/auth/DoctorRegisterPage.test.tsx](#)
- [web/src/pages/doctor/AgentOutputCard.test.tsx](#)
- [web/src/pages/doctor/ApprovalSupportingEvidence.test.tsx](#)
- [web/src/pages/doctor/ApprovalsPage.test.tsx](#)
- [web/src/pages/doctor/CasesPage.test.tsx](#)
- [web/src/pages/doctor/DoctorCalendarPage.test.tsx](#)
- [web/src/pages/doctor/DoctorDashboardPanel.test.tsx](#)
- [web/src/pages/doctor/DoctorFamiliesPage.test.tsx](#)
- [web/src/pages/doctor/DoctorFamilyDetailPage.test.tsx](#)
- [web/src/pages/doctor/DoctorMemberPage.test.tsx](#)
- [web/src/pages/doctor/DoctorProfilePage.test.tsx](#)
- [web/src/pages/doctor/PracticeProfileForm.test.tsx](#)
- [web/src/pages/doctor/approvalReview.test.ts](#)
- [web/src/pages/doctor/calendar/calendarUtils.test.ts](#)
- [web/src/pages/doctor/doctorSchedule.test.ts](#)
- [web/src/pages/doctor/memberClinicalTabs.test.tsx](#)
- [web/src/pages/doctor/reviewReasons.test.ts](#)
- [web/src/pages/doctor/safetyRules.test.ts](#)
- [web/src/pages/family/AppointmentsPage.test.tsx](#)
- [web/src/pages/family/FamilyDashboardPanel.adult.test.tsx](#)
- [web/src/pages/family/FamilyDashboardPanel.test.tsx](#)

- [web/src/pages/family/FamilyPage.test.tsx](#)
- [web/src/pages/family/FamilyRiskPage.test.tsx](#)
- [web/src/pages/family/HeadTransfer.test.tsx](#)
- [web/src/pages/family/IncomingInvitationsPanel.test.tsx](#)
- [web/src/pages/family/MyDoctorPage.test.tsx](#)
- [web/src/pages/family/OnboardingPage.test.tsx](#)
- [web/src/pages/family/PrivacyPage.test.tsx](#)
- [web/src/pages/notifications/NotificationsPage.test.tsx](#)
- [web/src/pages/profile/ProfilePage.test.tsx](#)
- [web/src/pages/public/PublicInformationPage.test.tsx](#)
- [web/src/pages/records/RecordsPage.test.tsx](#)
- [web/src/pages/records/VitalsPanel.test.tsx](#)
- [web/src/pages/records/extractionRefusal.test.ts](#)
- [web/src/pages/records/pollReading.test.ts](#)
- [web/src/pages/records/recordSummaryMeta.test.ts](#)
- [web/src/pages/records/vitalMeta.test.ts](#)
- [web/src/pages/triage/FamilyCaseProgress.test.tsx](#)
- [web/src/pages/triage/RequestPopup.test.tsx](#)
- [web/src/pages/triage/TriageJourney.test.tsx](#)
- [web/src/pages/triage/TriagePage.test.tsx](#)
- [web/src/routes/AppRouter.test.tsx](#)
- [web/src/routes/RouteGuard.test.tsx](#)
- [web/src/services/apiClient.session.test.ts](#)
- [web/src/store/slices/authSlice.test.ts](#)
- [web/src/styles/approvalVitalsGrid.test.ts](#)

Review notes: S4 Git authorship is confirmed for `doctor_queue_case_reference_test.dart` and `doctor_shell_after_login_test.dart`, but these two paths are absent from the ownership manifest. Request-hardening commit 7a90a77 is AI co-authored; its S1 scope does not make it an S1 student-authored commit. S2 ef35ed8 also changed historical backend `ExtractionAgentTests.cs` and `RecordServiceSortingTests.cs` (xUnit); those paths have since moved/changed and are not linked as current files.

Retained non-functional evidence links

- [ApacheBench raw local load output](#)
- [ZAP first-run/retest summary](#)
- [ZAP first report](#)
- [ZAP retest report](#)

Performance/security scripts or reports do not establish which student personally demonstrated them. Those fields remain blank.

